, the Bekenstein-Hawking Entropy, and the Running-Vacuum Form of Dark Energy from the Cosmological Horizon

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Date: April 2026
Status: DRAFT PRE-PRINT
Classification: Theoretical Physics / Gravitation / Cosmology

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Abstract

Applied to the cosmological horizon as a causal partition, the embedded-observation framework of [Main] determines the emergent action scale $\hbar = c^3\epsilon^2/(4G)$ from classical horizon temperature and thermal self-consistency, fixes the discreteness scale $\epsilon = 2\,l_p$ as the unique simultaneous solution, derives the Bekenstein-Hawking entropy $S = A/(4l_p^2)$ including the 1/4 coefficient by direct boundary-mode counting, and dissolves the cosmological constant problem by identifying the quantum vacuum energy and the effective cosmological constant as properties of logically distinct descriptions rather than commensurable quantities whose 10^{122} ratio requires cancellation. The recent LIGO/Virgo/KAGRA observation GW250114 (September 2025) confirms the black-hole area theorem at 99.999% confidence — a test of classical horizon dynamics, which the framework preserves; the derived 1/4 entropy coefficient itself has no direct empirical test. The framework predicts dynamical dark energy in Type II running-vacuum form as a *structural* prediction (functional form forced by boundary-entropy architecture; theorem-level magnitude $|\nu| \approx |\nu_{\rm QFT}| \sim 10^{-32}$, indistinguishable from zero) — consistent with the Bertini et al. (2025) RG-corrected ΛCDM fit to DESI DR2 + DES-Y5 + Planck 2018, which reports $\nu = -(2.5 \pm 1.3) \times 10^{-4}$ (consistent with zero at 2σ); a ~95% dark sector corollary in which the apparent invisible matter is a structural feature of the embedded observer's description rather than a new particle species; and a MOND-like acceleration scale $a_0 = cH/6$ from entropy displacement at the cosmological boundary, reproducing the baryonic Tully-Fisher relation and predicting Keplerian decline of galactic rotation curves at radii exceeding $r_M = \sqrt{6GM_B/(cH)}$ — the last confirmed by Jiao et al. (2023) in the Milky Way at ~ 17 kpc. The $H(z)$ dependence of a_0 predicts systematically more baryon-dominated rotation curves at high redshift, consistent with Genzel et al.'s detection of declining outer velocities at $z = 0.9$–2.4. All results follow from partition geometry alone and are independent of the specific substratum dynamics, treated in a separate companion paper [SM].

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1. Introduction

The cosmological observables that most resist explanation within standard physics — the value of $\hbar$, the Bekenstein-Hawking area law, the 10^{122} cosmological constant problem, the accelerated expansion, the ~95% dark sector, and the MOND-like behavior of galactic rotation curves — are treated here as consequences of a single structural fact: the cosmological horizon partitions the degrees of freedom of any sub-c observer into a visible sector and a causally inaccessible hidden sector.

Prerequisites. This paper takes as established the following results from a companion work [Main]: (i) an observer embedded in a deterministic bijection on a finite configuration space, coupled to a hidden sector that is non-trivial (C1), slow (C2: $\tau_S \ll \tau_B$), and high-capacity (C3: $N_H \gg N_V$), necessarily describes the visible sector using unitarily evolving quantum mechanics, and C1–C3 are necessary as well as sufficient (conditional on the eigenstate thermalization hypothesis for the C2 $\Leftarrow$ direction); (ii) the emergent description is uniquely determined by the partition, with any parameters of the emergent theory depending only on the partition boundary's geometric and thermodynamic properties (partition-relativity); (iii) the emergent transition probabilities $T_{ij}(t)$ uniquely determine the emergent Hamiltonian up to a diagonal-unitary D-gauge, via the phase-locking lemma applied to continuous-time sampling. The technical bridge from P-indivisible classical stochastic processes on finite configuration spaces to unitary quantum evolution on finite Hilbert spaces is Barandes' stochastic-quantum correspondence [35, 36], used as the companion paper's primary tool. We treat these results as inputs and apply them to the cosmological horizon. The cosmological horizon satisfies all three structural conditions: causal coupling via the Einstein constraint equations (C1), a slow-bath timescale of order the Hubble time $\tau_B \sim 10^{17}$ s (C2), and a hidden-sector capacity of order $S_{\mathrm{dS}} \sim 10^{122}$ modes (C3). The ETH condition for C2 necessity holds for the horizon complement (a generic chaotic many-body system), so the full equivalence applies in our universe.

The present paper reproves none of [Main]'s foundational results. Where a theorem below invokes a specific input — partition-relativity (§3.1), the emergent-description determination by transition probabilities (§3.2–§3.3), or the P-indivisibility/recurrence argument (§3.2 deep-sector lemma; §8 theorem requirements) — the invocation is stated as a known result and the reader is directed to [Main] for the proof.

The results are as follows. The emergent action scale $\hbar = c^3\epsilon^2/(4G)$ is determined by thermal self-consistency between the classical horizon temperature (computed from the Jacobson thermodynamic argument with no quantum input) and the Euclidean KMS period of the emergent QFT on the same horizon (§3). The discreteness scale $\epsilon = 2\,l_p$ is then fixed as the unique simultaneous solution of the action-scale relation and the definition of the Planck length (§4). The Bekenstein-Hawking entropy follows directly by mode counting, with the 1/4 coefficient derived rather than assumed as the ratio $2\pi/8\pi$ between the Euclidean KMS period and the Jacobson prefactor (§5); the recent LIGO/Virgo/KAGRA observation GW250114 confirms the area theorem at 99.999% confidence (the 1/4 coefficient itself has no direct test). The cosmological constant problem is dissolved by identifying the emergent quantum vacuum energy of order M_{Pl}^4 and the effective cosmological constant of order H^2/G as properties of logically distinct objects rather than commensurable quantities whose 10^{122} ratio requires cancellation (§6).

The framework's empirical content is developed in §7. Dynamical dark energy emerges in Type II running-vacuum form as a structural prediction forced by boundary-entropy architecture; the magnitude of the H^2 coefficient at theorem level is the emergent-QFT value $|\nu| \sim (M_{\mathrm{SM}}/M_{\mathrm{Pl}})^2 \sim 10^{-32}$, indistinguishable from zero. This is consistent with the Bertini et al. (2025) direct RG-corrected ΛCDM fit to DESI DR2 + DES-Y5 + Planck 2018, which reports ν consistent with zero at 2σ. The dark-sector corollary predicts that ~95% of the gravitational budget of the universe is invisible to the emergent QFT without requiring new fundamental particles — a concordance rather than a puzzle. The acceleration scale $a_0 = cH/6$ emerges from entropy displacement at the cosmological boundary, reproducing the MOND phenomenology in the deep-modified regime and predicting Keplerian decline at radii exceeding $r_M = \sqrt{6GM_B/(cH)}$; the latter is confirmed by the Jiao et al. (2023) detection of a Keplerian decline in the Milky Way rotation curve at ~ 17 kpc. The $H(z)$ dependence of a_0 predicts systematically more baryon-dominated rotation curves at high redshift, consistent with the declining outer velocities observed at $z = 0.9$–2.4 by Genzel et al.

Three epistemic tiers are worth distinguishing before the technical development begins. **Tier 1** comprises the results that follow from partition geometry alone — the derivation of $\hbar$, the BH area law with the 1/4 coefficient, the CC dissolution, and the Type II running-vacuum functional form of the dark energy. These do not depend on the specific substratum dynamics and would hold for any finite deterministic system satisfying the three structural conditions at its cosmological horizon. **Tier 2** comprises the dark-sector phenomenology: the ~95% invisible matter fraction, the MOND acceleration scale, and the rotation-curve crossover. These follow from the thermodynamic Clausius relation applied to the boundary reservoir and the geometric factor from spherical redistribution in de Sitter space. **Tier 3** comprises solution-specific quantities — the transition steepness between the MOND and Keplerian regimes, the greybody factor for Hawking radiation, and the specific microscopic realization of the substratum — which depend on the particular bijection φ and are addressed in the companion paper [SM].

The tier system maps to the framework's broader hierarchical structure articulated in [Structure §2]. *Tier 1* — the derivation of $\hbar$, the area law, the CC dissolution, and the Type II RVM functional form — operates at Level C $\times$ Level G4: these results hold for any partial-trace observational system on a horizon-like causal partition, not specifically for OI's cubic-lattice substratum. They are universality-class-level structural results. *Tier 2* — the dark-sector phenomenology — sits at Level D $\times$ Levels G1–G2, deriving from the thermodynamic Clausius relation applied to the specific boundary reservoir of OI's universality-class representative. *Tier 3* — solution-specific quantities — is at the bijection-specific level below Level D, depending on the particular φ within $[(S,\varphi)]/\mathcal{G}_{\mathrm{sub}}$. This stratification clarifies what a Tier 1 result claims: not "OI predicts $\hbar$ uniquely" but "any horizon-bounded embedded-observer system gets this $\hbar$, and OI provides one specific realization."

The paper is organized as follows. §2 identifies the cosmological horizon as a causal partition and verifies the structural conditions. §3 derives the emergent action scale $\hbar$ in four steps: the classical horizon temperature, boundary-only dependence, dimensional determination, and thermal self-consistency. §4 fixes the discreteness scale $\epsilon = 2\,l_p$. §5 derives the Bekenstein-Hawking entropy and discusses the GW250114 confirmation. §6 dissolves the cosmological constant problem. §7 presents the quantitative predictions for dark energy, the dark sector, and dark matter, with a summary table in §7.4. §8 discusses scope, the status of the discreteness scale, vacuum energy and the Higgs potential, and the framework's falsifiability conditions. §9 concludes. The cubic-lattice realization of the substratum — which fixes the Standard Model gauge structure via the wave equation — is developed in a companion paper [SM] and is independent of the GR results presented here. The reconstruction theorem and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S,φ) are developed in [Substratum].

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2. The Cosmological Horizon as Causal Partition

2.1 The partition

The cosmological horizon — the boundary beyond which no signal propagating at or below c can reach the observer — implements the causal partition of Lemma 2: Γ_V is the interior, Γ_H everything beyond. This is a consequence of GR's causal structure, not a modeling choice. The universality of the observed quantum mechanics follows from all sub-c observers sharing the same causal structure.

Different observers have different horizon areas (hence different S_{dS}), but $\hbar = c^3\epsilon^2/(4G)$ depends only on local geometric quantities — not on the observer's worldline or horizon area — so all observers share the same emergent action scale.

2.2 Verification of the conditions

(C1) In the ADM formulation, the bulk Hamiltonian is a sum of constraints that vanish on-shell; physical time evolution is generated by the boundary term at the horizon, which depends on data from both sides. The Hamiltonian and momentum constraints further correlate interior and exterior initial data on any Cauchy surface, so the physical phase space is a constraint surface within the kinematic product $\Gamma_V \times \Gamma_H$ (see §8.4 for implications). This is *stronger* than the $H_{\text{int}} \neq 0$ required by Lemma 2: constraints enforce correlations that persist on all timescales and cannot be perturbatively removed. **Satisfied.**

(C2) $\tau_B \gtrsim 1/H \sim 10^{17}$ s; for laboratory processes, $\tau_S \sim 10^{-15}$ s. Ratio: $\tau_S/\tau_B \sim 10^{-32}$. **Satisfied.**

(C3) Hidden sector has A/ϵ^2 modes with $A \sim 10^{52}$ m^2; visible sector has $\sim 10^{80}$ baryons. No laboratory process saturates the hidden sector. **Satisfied.**

2.3 Application

Lemma 1 is now a consequence of partition geometry: the horizon has finite area $A = 4\pi c^2/H^2$, bounding coupled modes to $A/\epsilon^2 < \infty$. The Part I theorem applies: the observer's reduced description is P-indivisible and, by either route of [Main, §3], equivalent to unitary QM with $\hbar$ determined by the partition boundary.

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3. The Emergent Action Scale $\hbar$

3.1 The classical horizon temperature

Jacobson [1] showed $dE = T\,dS$ applied to local causal horizons yields Einstein's equations, with $dE = (c^2\kappa/8\pi G)\,dA$, where κ is the surface gravity of the horizon. The entropy density is $\eta = 1/\epsilon^2$ — one coupled mode per minimal cell. This is not an assumption about the number of states per cell but a consequence of two results: ϵ is defined as the minimal distinguishable scale (Lemma 1), so each cell of area ϵ^2 contributes exactly one boundary mode that couples across the partition; the number of internal states per mode (the alphabet size q) is a gauge freedom with no observable consequences (the q-gauge theorem: two systems with different q produce identical transition probabilities if they share the same coupling structure). Thus $dS = dA/\epsilon^2$. Equating:

$$T_{\text{cl}} = \frac{c^2\epsilon^2\kappa}{8\pi G k_B}$$

For de Sitter ($\kappa = cH$): $k_B T_{\text{cl}} = c^3\epsilon^2 H/(8\pi G)$. No $\hbar$ appears.

Reconciliation with the internal component count. The counting $\eta = 1/\epsilon^2$ is one boundary mode per cell, whereas the substratum carries $K = 2d = 6$ internal components per site ([SM §4.5]). These are consistent, and the consistency is forced rather than assumed, once the components are read as what [SM §4.5, Theorem 6] establishes them to be: the $K = 2d$ components are the $2d$ link directions from a cell, with component a assigned to link direction $\hat{e}_a \in \{\pm\hat{e}_1, \dots, \pm\hat{e}_d\}$ and coupling only across that one link (the minimal-coupling factorization $\delta = 1$). A boundary cell on the horizon has exactly one outward-normal link crossing the partition; the other $2d - 1 = 5$ link directions are tangential to the boundary surface or inward, and carry no cross-partition coupling. The number of components whose coupling crosses the horizon per cell is therefore exactly one — the normal link — so $\kappa_m = 1$ and $\eta = 1/\epsilon^2$ follows from the link-direction structure, not from a separate mode-counting postulate. (The q-gauge argument settles the orthogonal question of states per mode; the component count is settled here by which links are normal.) This also confirms $\epsilon = 2\,l_p$ directly: with $\kappa_m = 1$ the relation $\epsilon = 2\sqrt{\kappa_m}\,l_p$ gives $\epsilon = 2\,l_p$, and the [SM §6] cutoff identification is unaffected. The Bekenstein-Hawking coefficient is independent of κ_m in any case — writing $\eta = \kappa_m/\epsilon^2$, the gap equation gives $\hbar = c^3/(4G\eta)$ and $S = \eta A = A/(4l_p^2)$ with κ_m cancelling — so the entropy law is a fixed point of the construction regardless, and the component count enters only the discreteness scale, where the link-direction structure fixes it at unity.

3.2 The emergent action scale

Step 1: Uniqueness. By partition-relativity (Prerequisites), $\hbar$ is derived, not free.

Step 2: $\hbar$ is structural, not volumetric.

Lemma (Boundary-only dependence). *Decompose the hidden sector as $\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$, where $\mathcal{C}_B$ denotes boundary modes (within the interaction range of H_{int}) and $\mathcal{C}_D$ denotes deep modes (the remainder of the hidden sector). If the classical Hamiltonian is spatially local, then on timescales $t \ll \tau_B$:*

$$T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$$

where $T_{ij}^{(B)}$ depends only on the V-B dynamics.

Proof. (i) *Coupling structure.* Spatial locality of the classical Hamiltonian implies the interaction decomposes as $H_{\text{tot}} = H_V + H_B + H_D + H_{VB} + H_{BD}$, with no direct V-D coupling: a deep hidden-sector mode must propagate through the boundary layer before influencing the visible sector. The coupling chain is $V \leftrightarrow B \leftrightarrow D$, not $V \leftrightarrow D$.

(ii) *Frozen deep sector.* Let Δ_D be the spectral gap of $H_D + H_{BD}$ restricted to the deep sector conditioned on a fixed boundary configuration. The deep-sector evolution over time t displaces any initial configuration by $\mathcal{O}(\Delta_D\, t)$ in phase space. The spectral gap gives the inverse relaxation time of the slowest mode, so $\Delta_D \le 1/\tau_B$ (with equality when the slowest mode dominates the relaxation). For $t \ll 1/\Delta_D \le \tau_B$: $\|\varphi_t^D - \mathrm{id}\| = \mathcal{O}(\Delta_D\, t) = \mathcal{O}(t/\tau_B)$.

(iii) *Factorization of the Liouville integral.* The transition probability is:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{b\in\mathcal{C}_B} \sum_{d\in\mathcal{C}_D} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))]$$

By (i), $\pi_V(\varphi_t(x_i, b, d))$ depends on d only through the back-reaction $B \leftarrow D$, which by (ii) shifts b by $\mathcal{O}(t/\tau_B)$. Expanding:

$$\pi_V(\varphi_t(x_i, b, d)) = \pi_V(\varphi_t^{VB}(x_i, b)) + \mathcal{O}(t/\tau_B)$$

where φ_t^{VB} is the flow restricted to $V \times B$ with D frozen. The d-sum then contributes $|\mathcal{C}_D|$ identical terms at leading order:

$$T_{ij}(t) = \underbrace{\frac{1}{|\mathcal{C}_B|} \sum_{b\in\mathcal{C}_B} \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i, b))]}_{T_{ij}^{(B)}(t)} + \mathcal{O}(t/\tau_B) \qquad \square$$

The correction is $\mathcal{O}(t/\tau_B) \sim 10^{-32}$ for laboratory processes. Since $T_{ij}(t)$ determines the emergent quantum description (Prerequisites; Barandes correspondence [35, 36]), and $T_{ij}^{(B)}(t)$ depends only on the V-B dynamics — which are characterized by the boundary geometry (A, ϵ, κ) and the constants c, G appearing in the classical Hamiltonian — the emergent action scale $\hbar$ inherits boundary-only dependence.

Corollary (deep-sector cardinality is gauge). *No observable of the emergent description — no transition probability, emergent Hamiltonian eigenvalue, or scattering amplitude — depends on the cardinality $|\mathcal{C}_D|$ of the deep hidden sector. Systems with the same $\mathcal{C}_V \times \mathcal{C}_B$ dynamics produce the same emergent physics to within $\mathcal{O}(\tau_S/\tau_B)$, whether $|\mathcal{C}_D|$ is finite, countably infinite, or uncountably infinite.*

Proof. $T_{ij}^{(B)}(t)$ is defined by a sum over $\mathcal{C}_B$ alone. The $\mathcal{C}_D$-dependent terms enter only through the $\mathcal{O}(t/\tau_B)$ back-reaction. Since the emergent Hamiltonian $\hat{H}_{\text{eff}}$ is uniquely determined by $\{T_{ij}(t)\}$ (phase-locking lemma, [Main, §3.1]; D-gauge theorem, §3.3), it inherits $\mathcal{C}_D$-independence at leading order. All observables — eigenvalues, transition amplitudes, scattering cross-sections — are functions of $\hat{H}_{\text{eff}}$ and therefore $\mathcal{C}_D$-independent. $\square$

The cardinality of the deep hidden sector is gauge in the same technical sense as the alphabet size q: just as two systems with different q are physically equivalent if they share the same coupling structure, two systems with different $|\mathcal{C}_D|$ are physically equivalent if they share the same $\mathcal{C}_V \times \mathcal{C}_B$ dynamics. Lemma 1 requires S finite for the recurrence proof ([Main, §2.3]); the corollary establishes that the deep sector's contribution to this finiteness has no observable consequences. The question "is the universe finite or infinite?" has no empirical content within the framework — it is identified as gauge.

Step 3: Dimensional determination. Step 2 excludes volumetric (deep-sector) quantities, leaving boundary quantities. The boundary carries both *local* geometric data (ϵ, κ, and the constants c, G of the classical Hamiltonian) and a *global* quantity: the total area A, which forms the dimensionless ratio $A/\epsilon^2 = S_{\rm dS}$. If $\hbar$ depended on $S_{\rm dS}$, it would be observer-dependent — different observers have different horizon areas (§2.1) — contradicting the universality of the emergent action scale. (All sub-c observers share the same local physics and hence the same $\hbar$; §2.1.) This excludes A, leaving the local boundary quantities c, G, ϵ, and κ. The surface gravity κ is excluded because it varies between observers and epochs while $\hbar$ is observed to be universal across observers and constant in time: a laboratory experiment measures the same $\hbar$ regardless of the observer's local cosmological-horizon parameters, and for the cosmological horizon $\kappa = cH$ is itself time-dependent ($\dot{H} \neq 0$). The unique combination of c, G, ϵ with dimensions of action:

$$\hbar = \beta \frac{c^3 \epsilon^2}{G}$$

Step 4: Thermal self-consistency. The classical substratum assigns the partition boundary a temperature $k_B T_{\rm cl} = c^2\epsilon^2\kappa/(8\pi G)$ (§3.1), containing no $\hbar$. The emergent QFT of Part I lives on this classical background, which has a bifurcate Killing horizon with surface gravity κ. Regularity of the Wick-rotated metric at the horizon requires Euclidean period $\beta = 2\pi c/\kappa$; any QFT on this background — including a lattice-regularized one — must therefore be periodic in imaginary time with the same period, giving a KMS state at temperature $T_Q = \hbar\kappa/(2\pi c k_B)$, with $\hbar$ unknown. This is a theorem *within* the derived QFT, not an external import. The two temperatures are computed independently — $T_{\rm cl}$ from the classical substratum alone (no QM), T_Q from the emergent QFT alone (no classical substratum details) — but they refer to the **same physical degrees of freedom**: the boundary modes $V \times B$ across which $H_{\rm int}$ couples the visible and hidden sectors. Step 2's boundary-only dependence shows that the emergent QFT is uniquely determined (at leading order in τ_S/τ_B) by these same $V \times B$ dynamics. Temperature is a state property; two complete descriptions of the same modes cannot assign different temperatures without logical contradiction. The matching $T_{\rm cl} = T_Q$ is therefore not an additional assumption but a consequence of the boundary modes being the same physical objects in both descriptions, with corrections at $\mathcal{O}(\tau_S/\tau_B) \sim 10^{-32}$:

$$\frac{c^2\epsilon^2\kappa}{8\pi G} = \frac{\hbar\kappa}{2\pi c}$$

κ cancels — a non-trivial cross-check on Step 3, which excluded κ from $\hbar$ on observer-universality grounds. The thermal matching here recovers that exclusion from independent physics: κ enters both sides of the consistency equation, and only the combinations of c, G, ϵ predicted by Step 3 survive. Solving:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}}$$

The derivation is a gap equation: ϵ is the free geometric input, $\hbar$ is the output. The match is non-trivial: $T_{\rm cl}$ could have depended on volumetric hidden-sector data (excluded by Step 2), or T_Q could have depended on the state (it does not — the KMS temperature is purely kinematic). That neither pathology obtains makes the gap equation a genuine determination. The non-circularity is structural: Part I establishes that a QFT emerges with *some* action scale $\hbar$; §3 determines *which* $\hbar$, using the independent classical temperature that Part I neither requires nor produces.

The counting $\eta = 1/\epsilon^2$ is definitional. Given that ϵ is the minimal distinguishable scale (Lemma 1), "one boundary mode per cell of area ϵ^2" is what "minimal" means: a cell finer than ϵ would subdivide the minimal scale, while counting fewer modes would require a distinction below ϵ to group them. The q-gauge theorem (alphabet size is observationally irrelevant) settles that what counts is the number of *cells*, not the number of internal states per cell. $\eta = 1/\epsilon^2$ is therefore not an independent physical ansatz about mode density but a consequence of the definition of ϵ combined with the irrelevance of sub-cell structure.

The predictive content lies not in the gap equation alone but in its consequences. The specific relationship $\hbar = c^3\epsilon^2/(4G)$ — rather than any other function of c, G, ϵ — produces: the Bekenstein-Hawking formula with the exact factor 1/4 (§5), the CC dissolution with $S_{\rm dS}$ as the compression ratio (§6.3), and the Type II running-vacuum form of the dark energy (§7.1). Any alternative $\hbar(\epsilon)$ would fail at least one of these checks. The situation is analogous to deriving the Schwarzschild metric with M as a free parameter: the derivation has genuine content (the functional form) even though one input is not determined from within.

Jacobson's original derivation [1] uses $\hbar$-containing forms; this paper does not — it uses the classical identity $dE = (c^2\kappa/8\pi G)\, dA$ and the classical entropy density $\eta = 1/\epsilon^2$. The logical ordering differs from Jacobson's: his argument derives G from η and $\hbar$; the present argument takes G as a parameter of the classical Hamiltonian (Lemma 3) and derives $\hbar$ from G and ϵ. The two are consistent — substituting $\eta = 1/\epsilon^2$ and $\hbar = c^3\epsilon^2/(4G)$ into Jacobson's formula $G_{\rm eff} = c^4/(8\pi\eta\cdot\hbar c)$ recovers $G_{\rm eff} = G$ — but the roles of input and output are reversed. The Gibbons-Hawking temperature [2] $k_B T_{\rm GH} = \hbar\kappa/(2\pi c)$ is recovered as a prediction, not used as an input. The KMS condition used above is derived for continuum QFT on bifurcate Killing horizons; for the lattice-regularized theory emerging from Part I, the relevant result is that thermal periodicity of the Euclidean Green's functions is robust against UV modifications of the dispersion relation, with corrections at $\mathcal{O}((\epsilon\kappa/c)^2)$ [3, 4]. For the cosmological horizon, $\epsilon\kappa/c = \epsilon H/c \sim 10^{-61}$, so lattice corrections are $\sim 10^{-122}$.

An independent necessity statement. The same conclusion is forced by an external theorem with a disjoint axiom base. For matter field equations whose principal polynomial is that of standard-model fields on a Lorentzian metric — the class in which the emergent sector of Part I falls, the single-cone structure being supported by the rotational-protection and species-universality results of [SM §3.1] — the gravitational closure theorem of Düll, Schuller, Stritzelberger and Wolz [Phys. Rev. D 97, 084036 (2018)] determines the gravitational action to be Einstein–Hilbert with cosmological constant, from canonical coevolution and diffeomorphism invariance alone, with no thermodynamic input. The derivation above and the closure theorem therefore force the same dynamics from disjoint axiom bases; the closure route leaves G and Λ as undetermined constants, which is precisely where the present relation $\hbar = c^3\epsilon^2/(4G)$ and the analysis of §6–7 take over. The closure theorem, moreover, is one member of a family of independent necessity statements the emergent sector inherits once its inputs stand: Lovelock's uniqueness theorem [J. Math. Phys. 12, 498 (1971)] — second-order, metric-only, generally covariant dynamics in four dimensions is Einstein–Hilbert with cosmological constant — and the spin-2 self-coupling and soft-graviton results [Weinberg, Phys. Rev. 138, B988 (1965); Deser, Gen. Rel. Grav. 1, 9 (1970)], by which a massless spin-two mode with a complete matter sector is forced into universal coupling and the Einstein–Hilbert form at low energy. Four routes — thermodynamic, closure, Lovelock, spin-2 bootstrap — with disjoint axiom bases converge on the same dynamics; this is also the structural answer to the standard analog-gravity objection that emergent spacetimes deliver kinematics but not Einstein dynamics: given the emergent cone, matter content, and spin-2 mode, the dynamics is over-determined rather than underived. Each route's hypotheses name an internal obligation the framework must independently meet (the single cone, the massless spin-2 bound state, the absence of extra light gravitational modes), and the perturbative members of the family carry their own known subtleties in the literature. The finite-ϵ deformation of the principal polynomial, and its comparison against the closure of weakly birefringent electrodynamics [arXiv:1708.03870], is an open consistency check.

3.3 Gauge-fixing and the dimensional obstruction

Steps 1–2 establish that $\hbar$ depends only on boundary quantities; Step 4 will fix its value via thermal matching. A prior question must be addressed: does the transition-probability data $\{T_{ij}(t)\}$ determine the emergent Hamiltonian uniquely, or does residual gauge freedom leave $\hbar$ underdetermined? The following theorem, completing the phase-locking argument (Prerequisites), shows the gauge freedom is physically trivial.

Theorem (D-gauge completeness). *Let $U(t) = e^{-iHt}$ with non-degenerate eigenvalues and non-vanishing configuration-basis overlaps. If $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ for all i, j, t, then $H' = DHD^\dagger$ for a time-independent diagonal unitary D.*

Proof. Write $U_{ij}(t) = \sum_a V_{ia}\, e^{-iE_a t}\, V_{ja}^*$ with $V_{ia} = \langle i|a\rangle$ and eigenvalues $\{E_a\}$, and similarly $U'_{ij}(t) = \sum_a V'_{ia}\, e^{-iE'_a t}\, V'^*_{ja}$.

Step 1 (Eigenvalue recovery). By the phase-locking argument of [Main, §3.1], $|U_{ij}(t)|^2 = |U'_{ij}(t)|^2$ for all t implies, via Fourier analysis, that the frequency sets $\{E_a - E_b\}$ and $\{E'_a - E'_b\}$ coincide. Non-degeneracy of energy gaps then gives $E'_a = E_{\sigma(a)} + E_0$ for some permutation σ and global shift E_0. The Fourier coefficients $a_{ij}^{kl} = V_{ik}V_{jk}^*V_{jl}V_{il}^*$ must match between primed and unprimed, which (by the non-vanishing condition on overlaps) forces $\sigma = \mathrm{id}$ up to relabeling. Thus $E'_a = E_a + E_0$.

Step 2 (Modulus recovery). From the diagonal Fourier coefficients $a_{ii}^{kl} = |V_{ik}|^2|V_{il}|^2$, the non-vanishing condition gives $|V'_{ia}| = |V_{ia}|$ for all i, a.

Step 3 (Phase structure). Write $V'_{ia} = V_{ia}\, e^{i\delta_{ia}}$. The off-diagonal Fourier coefficients require:

$$V'_{ik}V'^*_{jk}V'_{jl}V'^*_{il} = V_{ik}V_{jk}^*V_{jl}V_{il}^*$$

Substituting and canceling moduli:

$$e^{i(\delta_{ik}-\delta_{jk}-\delta_{il}+\delta_{jl})} = 1 \quad \text{for all } i, j, k, l$$

This is the double-difference condition: $\delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl} = 0 \pmod{2\pi}$. The general solution on the integer lattice is $\delta_{ia} = \alpha_i + \beta_a$ for phases $\{\alpha_i\}$, $\{\beta_a\}$. Thus $V' = D_\alpha\, V\, D_\beta$ where $D_\alpha = \mathrm{diag}(e^{i\alpha_i})$ and $D_\beta = \mathrm{diag}(e^{i\beta_a})$. The D_β factor is absorbed into the eigenvector phase convention (physically irrelevant), giving $H' = D_\alpha H D_\alpha^\dagger$ with $D = D_\alpha$ a time-independent diagonal unitary. $\square$

The D-gauge is physically trivial (basis rephasing). Cross-interval transition probabilities fix the relative gauge up to a single global phase; in the continuum limit this becomes an unobservable smooth function $e^{i\theta(t)}$.

Dimensional obstruction. The unitary $U(t)$ is dimensionless; $\hbar$ enters only via $\hat{H} = i\hbar\, \partial_t U \cdot U^\dagger$. No dimensionless data can fix a dimensionful constant — Step 4's thermal self-consistency provides the necessary dimensionful input.

Remark (phase recovery under discrete sampling). The spatial lattice discreteness does not discretize time: the continuous-time extension ([Main, §2.3]) provides $T_{ij}(t)$ as a genuinely continuous function of t via the Liouville marginalization of the Hamiltonian flow (Lemma 3). The D-gauge theorem therefore applies to the continuous object without aliasing corrections. Aliasing would arise only if the observer sampled T_{ij} at discrete intervals, in which case trans-Planckian frequencies could be misidentified — but these lie outside the emergent QFT's domain. The phase-recovery error is controlled by the same

$\mathcal{O}(\tau_S/\tau_B) \sim 10^{-32}$ suppression that governs the boundary-only dependence of §3.2.

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4. Self-Consistency and the Discreteness Scale

Rearranging $\hbar = c^3\epsilon^2/(4G)$:

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\,l_p^2$$

The framework yields $\epsilon = 2\,l_p$ exactly. This is algebraically equivalent to the formula for $\hbar$: the framework contains one free geometric parameter (ϵ), and thermal self-consistency establishes its relationship to $\hbar$. The result is a self-consistency condition constraining ϵ to the Planck regime, not an independent determination. If $\epsilon^2 \ll l_p^2$, sub-Planckian subcells would create a second trace-out, breaking the single-valuedness of $\hbar$. If $\epsilon^2 \gg l_p^2$, the emergent description would assign distinct quantum states to physically identical configurations. The framework's predictive content comes from the relationship $\hbar = c^3\epsilon^2/(4G)$, the Bekenstein-Hawking formula, and the consequences that follow.

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5. The Bekenstein-Hawking Entropy

With $\epsilon = 2\,l_p$, the number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\,l_p^2} = S_{\text{dS}}$$

where $S_{\text{dS}} = A/(4\,l_p^2)$ is the Bekenstein-Hawking entropy of the de Sitter horizon. The factor of $1/4$ in the Bekenstein-Hawking formula — historically introduced as a proportionality constant — is here derived: each minimal cell of area $\epsilon^2 = 4\,l_p^2$ on the horizon contributes one unit of entropy, and $A/(4\,l_p^2) = A/\epsilon^2$ recovers the standard formula exactly. The factor of 4 in $\epsilon^2 = 4l_p^2$ traces directly to §3 Step 4's thermal matching: the ratio $2\pi/8\pi = 1/4$ between the Euclidean KMS period (numerator: thermal periodicity of QFT on a Killing horizon) and Jacobson's Einstein-equation prefactor (denominator: gravity's coupling in $dE = (c^2\kappa/8\pi G)dA$). The Bekenstein-Hawking $1/4$ is therefore not an arbitrary constant but the consistency condition between gravitational and quantum thermal physics. The same minimal-cell counting argument applies to any horizon with area A, giving $S = A/(4l_p^2)$ for black-hole horizons as well; the recent observational confirmation of the area law on GW250114 [5] is consistent with this extension.

Observational confirmation: GW250114. The Bekenstein-Hawking area law $dA \geq 0$ has long been regarded as a theoretical prediction of black-hole thermodynamics, but until 2025 the empirical evidence came primarily from agreement between classical horizon dynamics and semiclassical Hawking radiation predictions — none of it testing the 1/4 coefficient directly. The LIGO/Virgo/KAGRA observation of GW250114 (September 2025) provides the first direct test of the area law at the percent level: comparing the remnant horizon area inferred from the ringdown to the sum of initial horizon areas inferred from the inspiral, the area increase is consistent with the theorem at 99.999% confidence. The scope of this test is important to state precisely: GW250114 compares horizon areas and does not measure entropy, so it tests Hawking's area theorem — classical horizon dynamics, which the framework preserves — and not the 1/4 entropy coefficient. The coefficient, fixed here as the exact ratio $2\pi/8\pi$ between Euclidean KMS periodicity (thermal physics) and Jacobson's Einstein-equation prefactor (gravitational coupling), is a derived structural constant with no direct empirical test at present; any framework retaining classical general relativity passes the GW250114 test identically, so the observation is consistent with, but non-discriminating for, the framework. Untested is not untestable: the coefficient is equivalent to the Hawking-temperature normalization ($T = \hbar\kappa/2\pi c k_B$ plus the first law fixes it), so any measurement of a horizon's radiation would fix it directly. No known astrophysical source is accessible — a solar-mass black hole radiates at ~ 60 nK against the 2.7 K CMB — but one physically contingent channel exists: the final evaporation burst of a primordial black hole ($M_i \sim 5 \times 10^{14}$ g expiring today), whose burst light curve would measure $T(M)$ and hence the coefficient. Dedicated all-sky burst searches are active — Fermi-LAT [45], HAWC [46], and most recently LHAASO's all-sky search [47] — with only rate-density limits so far, and next-generation sensitivity projected for CTAO [48]; detection is contingent on PBHs existing in the relevant mass window. Analogue-gravity horizon experiments measure the kinematic (2π/KMS) side of the ratio but not the gravitational ($8\pi G$) side. The ongoing LVK observing runs will accumulate additional events testing the area theorem across a range of progenitor masses and spins.

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6. Dissolution of the Cosmological Constant Problem

6.1 The two levels of description

From finite-dimensional QM to QFT. Part I delivers QM on a finite-dimensional Hilbert space. The $N = A/\epsilon^2$ cells discretize the visible sector; spatial locality of the classical substratum restricts which cells interact on any given timescale. The emergent Hilbert space decomposes as $\mathcal{H} = \bigotimes_k \mathcal{H}_k$ — a lattice-regularized QFT with UV cutoff at $\epsilon = 2\,l_p$. The tensor product structure follows from the spatial product structure of the classical configuration space $\mathcal{C}_V = \mathcal{C}_1 \times \cdots \times \mathcal{C}_N$ (one factor per cell), which is not required by Lemma

1 but is a consequence of spatial locality in the classical substratum: each cell's configuration is a local degree of freedom.

Locality preservation theorem. *If the classical Hamiltonian is spatially local (couples only neighbors), then the emergent quantum Hamiltonian inherits spatial locality.*

Proof. For infinitesimal dt and $x' \neq x$: $T_{x,x'}(dt) = |\langle x'|\hat{H}|x\rangle|^2\, dt^2 + O(dt^3)$. If x, x' differ at non-neighboring sites, spatial locality of the classical dynamics gives $T_{x,x'}(dt) = 0$ at leading order, hence $|\langle x'|\hat{H}|x\rangle|^2 = 0$. By the D-gauge theorem (§3.3), the residual phase freedom is a diagonal unitary D that preserves the vanishing of off-diagonal moduli, so $\langle x'|\hat{H}|x\rangle = 0$ in any gauge. The emergent Hamiltonian has the form $\hat{H} = \sum_k \hat{h}_k + \sum_{\langle k,l\rangle} \hat{h}_{kl}$. $\square$

Scope of the emergent QFT. The CC dissolution requires only that the emergent description assigns zero-point energy $\sim \hbar\omega/2$ per mode with UV cutoff at ϵ^{-1} — a property shared by any lattice QFT with local couplings, regardless of gauge group or matter content. The framework's predictions in Part II ($\hbar$, S_{dS}, the Type II RVM form of the dark energy) depend on partition geometry, not on the emergent field content.

Classical substratum (what geometric measurements probe): The horizon has classical thermal equilibrium. By the Friedmann equation, $\rho_{\text{crit}} = 3H^2c^2/(8\pi G)$. No zero-point energy, no discrepancy. Total vacuum energy density: $\rho \sim H^2/G \sim 10^{-9}$ J/m^3 — precisely observed.

Emergent QFT (what local quantum measurements probe): Zero-point energy $\frac{1}{2}\hbar\omega$ per mode summed to the Planck cutoff gives $\rho_{\text{QFT}} \sim 10^{113}$ J/m^3 — a $\sim 10^{122}$ discrepancy with the observed value [6, 7, 8].

6.2 Why gravity sees the classical value

Spacetime geometry is part of the classical substratum (Lemma 2): the metric evolves via Einstein's equations *before* the trace-out. The stress-energy sourcing gravity is the classical stress-energy of the total microstate, not the expectation value of an emergent quantum operator. The semiclassical equation $G_{\mu\nu} = 8\pi G\langle\hat{T}_{\mu\nu}\rangle$ is itself an emergent approximation; at the classical level, the gravitational field equations never encounter the zero-point sum.

Ontological commitment. Classical spacetime is logically prior to QM; QM is emergent. This ordering is not a claim that spacetime is a substance-level primitive of the framework — the framework has no substance-level primitives, and under the reconstruction theorem, spacetime emerges from $[(S,\varphi)]/\mathcal{G}_{\text{sub}}$ given the framework's irreducible starting commitments (the empirical fact that observation occurs, the empirical inputs E1–E4, and the structural assumptions A1–A6). It is a claim about the direction of the derivation chain — that the trace-out producing QM presupposes the classical geometric structure, not the reverse. This ordering is forced by the logical structure of the derivation through three independent requirements:

(i) Definiteness. The trace-out that produces QM is defined by a *specific* partition $\Gamma = \Gamma_V \times \Gamma_H$. The partition must be definite — not in superposition, not subject to quantum uncertainty — for the marginalization integral ([Main, §1.4]) to be well-defined. A partition in superposition would yield an incoherent mixture of inequivalent quantum theories, not a single QM. The metric at the partition boundary must therefore be classical.

(ii) Non-circularity. The partition is defined by the causal structure: Γ_H is the set of degrees of freedom beyond the observer's causal reach (§2.1). Causal structure is determined by null geodesics of the metric. If the metric were derived from QM, then the derivation would require: QM $\rightarrow$ metric $\rightarrow$ causal structure $\rightarrow$ partition $\rightarrow$ QM — a circular dependency with no entry point. The circle breaks only if the metric exists prior to the partition.

(iii) Uniqueness of $\hbar$. The emergent action scale is determined by the boundary geometry (§3.2). If the geometry were itself quantum-mechanical, $\hbar$ would depend on a quantum state — but $\hbar$ is state-independent by observation. The geometry-first ordering is the only one consistent with a universal $\hbar$.

Any framework in which QM is logically prior to the metric must either (a) treat the zero-point sum as a gravitational source — producing the 10^{122} discrepancy requiring cancellation or fine-tuning — or (b) invoke an independent mechanism to decouple quantum vacuum energy from gravity. In the present framework, the zero-point sum is an artifact of the emergent description and never enters the stress-energy tensor; no decoupling mechanism is needed.

Cumulative evidence for the ordering. The geometry-first ordering produces a set of consequences that match observation, none of which were designed into the starting point. They are listed with their dependencies stated: items (2) and (3) are one algebraic fact, item (4) is partly definitional, and item (7) is a one-parameter derivation plus a three-parameter match, so eight of the entries are mutually independent: (1) the CC dissolution — the observed vacuum energy sits at $\rho \sim H^2/G$, the classical baseline, without fine-tuning (§6.3); (2)–(3) the relation $\hbar = c^3\epsilon^2/(4G)$ with its 1/4 coefficient and the Bekenstein-Hawking formula $S = A/(4l_p^2)$ — *one* consequence, since $\epsilon = 2\,l_p$ is algebraically equivalent to the gap equation (§4) and the entropy formula is its re-expression through the mode count $S = A/\epsilon^2$; the independent content is the single functional form and the KMS/Jacobson ratio behind the 1/4 (§5); (4) the dark-sector concordance — ~95% of ρ_{crit} is invisible to QFT; the *fraction* is $1-\Omega_B$ by construction (from $S_{\text{struct}}/S_{\text{total}} = \Omega_B$, §7.3), so it is partly definitional rather than an independent match — the non-trivial content is the matter-like vs dark-energy-like *split* within that 95%, tested separately (§7.1–7.3); (5) dark energy in running-vacuum form — the Type II functional form forced by boundary-entropy architecture, with theorem-level magnitude $|\nu| \approx |\nu_{\text{QFT}}| \sim 10^{-32}$ (observationally indistinguishable from ΛCDM), consistent at 2σ with the Bertini et al. (2025) RG-corrected ΛCDM fit (itself consistent with zero); DESI DR2's preference for evolving dark energy favours the running-vacuum *form* over plain ΛCDM at the qualitative level, but its phantom-crossing *magnitude* is in mild tension (3–4σ) with the framework's $\nu \approx 0$ prediction (§7.1) [15]; (6) the MOND acceleration scale $a_0 = cH/6$ and the baryonic Tully-Fisher relation, derived from entropy displacement (§7.3); (7) the SM gauge group (a discrete structural output), together with the coupling sector's derived content — the induced $1/\alpha_0 = 23.25$ and the universality of the threshold — the three coupling values at M_Z being reproduced through three matched parameters (δ_0, A, B), so that the $< 0.1\%$ figures are fits rather than consequences [SM §6]; (8) declining rotation curves at high redshift, confirmed by Genzel et al. (§7.3); (9) $\bar{\theta} = 0$ at all energy scales; (10) deep-MOND cluster dynamics (e.g. Coma at the virial radius) matched by the simple interpolation function — though rich-cluster cores retain an open 1.0–1.5× residual not closable by baryons or by the framework's predicted light neutrinos (§7.3); (11) the Bullet Cluster lensing morphology reproduced by frozen boundary entropy (§7.3). Quantum-first orderings produce the 10^{122} discrepancy, have no natural account of the dark sector, and treat $\hbar$ and the $1/4$ factor as unexplained inputs. The cumulative case is not a single tiebreaker but a pattern: the quantitative outputs of the geometry-first ordering are consistent with observation — with the DESI phantom-crossing magnitude (item 5) carried as the one live 3–4σ tension on its own dated schedule — while the central quantitative output of the quantum-first ordering is the worst prediction in physics.

The classical vacuum energy scale. The framework does not explain *why* $\rho \sim H^2/G$; this is set by initial conditions via the Friedmann equation. What it does is reclassify the CC puzzle: from quantum vacuum cancellation to classical initial conditions. The two problems are not equivalent. The QFT CC problem requires a cancellation between independent contributions — the bare cosmological constant and the zero-point energies of every field — accurate to 122 decimal places, re-tuned at each order in perturbation theory. The initial-conditions question asks why a single parameter (Λ, or equivalently H) takes its observed value; the scale $\rho \sim H^2/G$ is the natural energy density of classical cosmology, not a fine-tuned cancellation between unrelated quantities. The framework eliminates the 122-digit cancellation entirely; the remaining question — why H takes its value — is a standard cosmological problem addressable by inflationary dynamics or other initial-conditions mechanisms.

Quantum corrections to gravity. The emergent quantum description does feed back into gravitational dynamics through state-level quantities: §7.1 derives dark energy evolution because the emergent vacuum energy depends on H through the hidden sector's volume. What the framework excludes is the zero-point sum as a gravitational source. The zero-point energy is the ground-state eigenvalue of the emergent Hamiltonian $\hat{H}_{\text{eff}}$, which acts on $\mathcal{H}_V$. But gravity is sourced by the classical Hamiltonian H_{tot}, which acts on the full phase space Γ — and $\hat{H}_{\text{eff}}$ and H_{tot} are not the same operator. The eigenvalues of $\hat{H}_{\text{eff}}$ (including the zero-point energy) are properties of the *emergent description*; the energy that H_{tot} assigns to the corresponding classical microstates is the *classical vac-*

uum energy $\rho \sim H^2/G$. The Equivalence Principle applies to H_{tot}: everything that contributes to H_{tot} gravitates. The zero-point sum does not contribute to H_{tot}; it appears when the classical transition probabilities are reorganized into the Hilbert space formalism ([Main, §3.1]), not when the classical dynamics is evolved. The structural/volumetric distinction of §3.2 captures this: $\hbar$ (structural, a property of the correspondence) does not gravitate; the vacuum energy (volumetric, state-dependent, a property of H_{tot}) does, but at the classical scale $\rho \sim H^2/G$ rather than $\rho \sim M_{\text{Pl}}^4$.

6.3 The structural origin of the 10^{122} discrepancy

$\rho_{\text{QFT}}/\rho_{\text{classical}} \sim M_{\text{Pl}}^4/(H^2/G) = S_{\text{dS}} = A/(4\,l_p^2)$ — the Bekenstein-Hawking entropy, now identified as the information compression ratio of the trace-out. The "worst prediction in physics" is the entropy of the observer's blind spot — a category error, not a fine-tuning failure. Wolpert's limits of inference [9] confirm this cannot be resolved from within: both geometric and quantum measurements are faithful to their respective descriptions, and no embedded observer can determine which is more fundamental. This is not a prediction awaiting future data — the observed vacuum energy sits at the classical geometric scale $\rho \sim H^2/G$, the value the framework expects.

Convergence with Maldacena's de Sitter observer cancellation. Maldacena [38] has recently shown that the unphysical phase of the de Sitter sphere partition function — long understood to obstruct a clean dS thermodynamic interpretation — cancels exactly when one incorporates an observer with a clock. The framework presented here arrives at the same cancellation by a different route: the substratum-count formulation makes the partition trace-out the *defining* operation rather than an added ingredient, so the cancellation Maldacena requires an observer-with-clock to achieve is structurally forced. CLPW [39] establish that the von Neumann algebra of QFT observables in a gravitational subregion promotes from type III to type II once an observer is included, with the resulting entropy pinned to the substratum's $\log|G_{\text{sub}}|$ in the present construction. HUZ [40] argue that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom; the present framework provides a microscopic realization in which that dimension is computed directly from the partition geometry. The post-2022 mainstream observer-essentiality literature thus converges with the present construction on substantive content while differing in mechanism — a triangulation that strengthens the empirical case for both programs.

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7. Predictions and Tests

7.1 Dark energy evolution in running vacuum form

$\hbar$ is H-independent (§3.2), but the emergent vacuum energy is a state-level quantity depending on the hidden sector's volume. Any smooth $\rho_\Lambda(H)$ admits the Taylor expansion

$$\Lambda_{\text{eff}}(H) = \Lambda_0 + \frac{3\nu}{8\pi G}(H^2 - H_0^2) + \mathcal{O}((H^2 - H_0^2)^2)$$

This is the Running Vacuum Model [10, 11]. The framework forces this structural form through three architectural commitments: (i) the boundary entropy $S_{\text{dS}} = A/\epsilon^2$ depends on H via $A = 4\pi c^2/H^2$, so the emergent vacuum energy inherits an H-dependence; (ii) matter is conserved in the trace-out ontology (§7.2), which forbids direct vacuum–matter energy exchange; (iii) Bianchi consistency with matter conservation then forces $G(H)$ to run in tandem with $\rho_\Lambda(H)$ [11]. The functional form is therefore a structural prediction: the framework commits to Type II RVM.

Type II identification. The architecture just described places the framework in the *Type II* class of RVM implementations [11]. Matter is locally conserved, $G(H)$ runs mildly with cosmic expansion, and the Bianchi identity is preserved by the combined running of $\rho_{\text{vac}}(H)$ and $G(H)$. By contrast, *Type I* RVM implementations assume $G = G_N$ fixed and introduce a direct vacuum–DM exchange term $Q = 3\nu H\rho_{\text{dm}}$; these are physically distinct models sharing the same functional form for $\rho_{\text{vac}}(H)$.

What the framework determines about ν. The architecture above fixes the functional form but not the coefficient. A rigorous derivation of ν requires completing §7.3's dark-energy / dark-matter split analysis — specifically, the H-dependence of the uniform-vs-structured partition of boundary entropy — which has not been done at the level of a theorem. What the foundations *do* determine are:

- $\rho_s(H) = 3c^2H^2/(8\pi G) = \rho_{\text{crit}}(H)$ exactly, from the gap equation combined with classical-horizon thermodynamics (§7.2). The total gravitating boundary energy tracks the critical density at every H as an identity, not a prediction.
- $S_{\text{struct}}/S_{\text{total}} = \Omega_B(H)$, from the Clausius-relation entropy-displacement argument of §7.3. The structured (matter-associated) fraction of boundary entropy equals the baryon fraction at each epoch.
- What is *not* fixed: how the non-baryonic $(1 - \Omega_B)$ fraction further subdivides into the cosmological-constant-like (uniform) and dark-matter-like (structured) components as a function of H. That subdivision determines ν.

Magnitude at theorem level. The self-consistent emergent-QFT calculation gives the H^2 coefficient directly. In OI the matter fields and the curved-spacetime background are both emergent from the classical discrete substratum; the Shapiro-Solà -function formula [33, 34] for the vacuum-energy running coefficient

$$\nu = \frac{1}{48\pi^2}\sum_i a_i \frac{M_i^2}{M_{\rm Pl}^2}$$

is derived from one-loop QFT on a curved semiclassical background and applies rigorously within the emergent QFT sector, with M_i the physical masses of the emergent fields. Applied to SM content ($M_i \leq m_t \approx 173$ GeV), the formula gives

$$|\nu_{\rm QFT}| \sim (M_{\rm SM}/M_{\rm Pl})^2 \sim 10^{-32}$$

— observationally indistinguishable from zero.

A potential additional contribution from the substratum mode count is constrained by the boundary architecture established in §3.1 and §5: boundary modes are catalogued by spatial location alone (one degree of freedom per cell of area ϵ^2), with no inherent frequency assignment. Under the natural 2D dispersion $dN/d\omega \propto \omega$, modes concentrate near the UV cutoff and contribute a fraction $(H\epsilon)^2 \sim 10^{-122}$ at horizon frequency, well below the emergent-QFT magnitude. Absent additional substratum structure giving log-uniform boundary-mode distribution at the cosmological horizon, the framework's prediction reduces to $|\nu| \approx |\nu_{\rm QFT}|$ — observationally indistinguishable from zero at all foreseeable precision, including DESI Year 5.

Observational status. DESI data [12, 13] report evolving dark energy at 2.8σ–4.2σ. The 2026 multi-variant analysis [14] of DESI DR2 + Planck PR4 + supernovae tests several Type I RVM variants (vacuum–DM exchange), finding the "flipped RVM" and "RVM with threshold" variants preferred over ΛCDM at a level comparable to the more flexible $w_0 w_a$CDM parameterization — the running-vacuum functional form this framework predicts is favoured by the data at the qualitative level, though the Type I variants tested are not a direct test of OI's Type II prediction.

A direct Type-II-equivalent test is provided by Bertini, Novaes, von Marttens, and Shapiro (arXiv:2509.24026) [15], who fit a renormalization-group-corrected ΛCDM model — structurally equivalent to Type II RVM at leading order in ν, with both $G(\mu)$ and $\Lambda(\mu)$ running and matter conserved — against Planck 2018 + DESI DR2 + DES-Y5, reporting

$$\nu^{\rm Bertini} = -(2.5 \pm 1.3) \times 10^{-4}$$

with the ΛCDM limit at the 2σ boundary of the posterior. This is consistent at 2σ with the framework's prediction ($\nu \approx 0$). An earlier pre-DESI canonical RVM fit by Gómez-Valent and Solà [16] returned $\nu = (1.34 \pm 0.38) \times 10^{-3}$ with 3.5σ preference over ΛCDM; the migration from this value to Bertini's as the data improved from pre-DESI to DESI DR2 is itself informative. DESI Year 5

($\sim 1\%$ precision on ν) provides the decisive test: a null at that precision confirms the prediction; a high-confidence detection of $|\nu|$ at the 10^{-4} level would require additional substratum structure not currently derivable from §3-§5.

Coupled neutrino test. The same dataset that constrains ν also constrains the neutrino sector. The DESI DR2 + CMB analysis [17] finds $\Sigma m_\nu < 0.0642$ eV (95% CL) assuming ΛCDM, prefers normal mass ordering, and bounds the lightest neutrino mass at $m_l < 0.023$ eV — all three results match the OI predictions of normal ordering with Σm_ν near the oscillation minimum of 0.059 eV. The same analysis finds a 3σ tension with the lower oscillation limit assuming ΛCDM, interpreted as "a hint of new physics not necessarily related to neutrinos" — a mismatch the framework's RVM dark energy would *relax*: an evolving-DE analysis loosens the ΛCDM Σm_ν bound, comfortably accommodating the framework's ~ 0.059 eV prediction, though the RVM-corrected bound has not yet been computed. JUNO first results [18, 19] confirm the world-leading precision on $\sin^2\theta_{12} = 0.3092 \pm 0.0087$ and $\delta m^2 = (7.48 \pm 0.10) \times 10^{-5}$ eV^2, with the solar/reactor 1.5σ tension persisting. The neutrino sector predictions and the RVM dark energy prediction are coupled in the framework — both follow from the same lattice structure — and the joint observational support is therefore not independent confirmation of separate effects but a single empirical pattern consistent with the framework's central mechanism.

Falsification gradient and decisive test structure. The framework predicts $w(z) \approx -1$ structurally — the running of the cosmological constant in Type II RVM is H^2-suppressed relative to the leading Λ contribution, so the effective equation of state stays at -1 to within deviations of order ν. DESI DR2's mild preference for phantom crossing ($w_0 > -1$ at present, $w_a < 0$ with $w(z > z_{\text{cross}}) < -1$) is therefore in mild tension (3-4σ) with the framework's prediction. The Bertini et al. [15] RG-corrected ΛCDM fit, which is the closest published Type II RVM equivalent, sits at the 2σ boundary of the posterior — consistent with the framework but not yet a confirmation. DESI Year 5 ($\sim 1\%$ precision on ν, available 2028-2029) provides the decisive three-way discrimination: (i) a null result at that precision *confirms* the framework's prediction $\nu \approx 0$ and disconfirms the DESI DR2 phantom-crossing preference; (ii) a high-confidence detection of $|\nu| \sim 10^{-4}$ confirms running-vacuum dynamics consistent with both Type II RVM structure and DESI DR2 trends, but at a magnitude requiring substratum structure beyond §3-§5; (iii) a high-confidence detection of phantom crossing falsifies the framework's prediction at structural level. The current mild tension with DESI DR2 is therefore the kind of healthy empirical exposure — a falsification gradient with concrete decisive test scheduled — that distinguishes a framework engaged with data from one tuned to fit it.

7.2 The dark-sector corollary

The trace-out that produces QM has an automatic gravitational consequence.

Corollary (Invisible gravitational budget). *Under Lemmas 1–3 and con-*

ditions (C1)–(C3), the cosmological trace-out that produces quantum mechanics simultaneously renders ~95% of the universe's gravitational budget invisible to the emergent QFT.

Proof. Three steps, each using only results already established.

(i) Total effective density. The horizon carries $S_{\mathrm{dS}} = A/\epsilon^2$ modes (§5). The classical temperature is $T_{\mathrm{cl}} = c^3\epsilon^2 H/(8\pi G k_B)$ (§3.1), computed entirely within the classical substratum with no reference to $\hbar$ or the emergent quantum description. The total thermal energy is $E_s = S_{\mathrm{dS}} \cdot k_B T_{\mathrm{cl}}$. By the shell theorem, distributing over $V_H = 4\pi c^3/(3H^3)$:

$$\rho_s = \frac{E_s}{V_H} = \frac{A}{\epsilon^2} \cdot \frac{c^3\epsilon^2 H}{8\pi G} \cdot \frac{3H^3}{4\pi c^3} = \frac{3c^2H^2}{8\pi G} = \rho_{\mathrm{crit}}$$

The identity $\rho_s = \rho_{\mathrm{crit}}$ is a consequence of Jacobson's Clausius relation applied to the de Sitter horizon. Verlinde [20] takes the identity as a premise; the present framework grounds it in the Clausius relation of the underlying classical substratum, using the boundary-mode counting fixed in §3.1. The factor ϵ^2 cancels between $S_{\mathrm{dS}} \propto 1/\epsilon^2$ and $T_{\mathrm{cl}} \propto \epsilon^2$, so the result is independent of the specific value of ϵ: what the identity actually uses is Jacobson's $dE = (c^2\kappa/8\pi G)\,dA$ plus an area-law entropy plus the Friedmann volume.

Three clarifications are essential for what follows.

Classical, not emergent. The calculation above uses only pre-trace-out quantities: the number of boundary modes (A/ϵ^2), the classical temperature (T_{cl}), and the Hubble volume (V_H). No step invokes the emergent quantum description. This is what distinguishes ρ_s from the QFT zero-point energy $\rho_{\mathrm{QFT}} \sim M_{\mathrm{Pl}}^4$: the latter is computed as $\sum \frac{1}{2}\hbar\omega$ over modes of the *emergent* quantum field theory — a quantity that exists only after the trace-out. The framework denies that ρ_{QFT} gravitates (§6.2), because it is an artifact of the emergent description. The boundary entropy's thermal energy, by contrast, is a classical quantity that predates the trace-out and gravitates at the classical level. That both involve $\hbar$ numerically (because $T_{\mathrm{cl}} = T_{\mathrm{GH}}$) is a consequence of the gap equation's self-consistency, not a dependence on QM. The ratio $\rho_{\mathrm{QFT}}/\rho_s \sim S_{\mathrm{dS}} \sim 10^{122}$ — the same compression ratio identified in §6.3 — confirming that the two quantities occupy different levels of the framework's hierarchy.

Physical locus. The entropy modes are physically localized on the horizon — they are the hidden-sector boundary degrees of freedom of §3.2. The uniform-medium description is a gravitational equivalence (shell theorem), not a claim about physical delocalization. No entropy occupies the bulk.

Why the equivalence is useful. When matter is present, it locally deforms the horizon geometry, redistributing the boundary entropy. The gravitational effect of this redistribution on bulk observers is most easily computed using the uniform-medium equivalence: the deformed boundary is equivalent to a non-uniform effective bulk source.

General d. Running the same construction in d spatial dimensions: the horizon is a $(d-1)$-sphere of area $A_{d-1} = [2\pi^{d/2}/\Gamma(d/2)]r_H^{d-1}$; the Hubble volume is $V_d = [\pi^{d/2}/\Gamma(d/2+1)]r_H^d = A_{d-1}r_H/d$; the classical horizon temperature $T_{\rm cl} \propto H$ is dimension-independent (Gibbons–Hawking); and the Friedmann equation in d spatial dimensions is $H^2 = [16\pi G_d/(d(d-1))]\rho$. The boundary entropy is $S = A_{d-1}/\epsilon^{d-1}$ (one mode per ϵ^{d-1} cell) and the gap equation fixes $\hbar = c^3\epsilon^{d-1}/[2(d-1)G_d]$ by the same thermal self-consistency argument as at $d=3$. Collecting the geometric factors in $\rho_s = S\,k_B T_{\rm cl}/V_d$ and dividing by $\rho_{\rm crit}$ gives

$$\frac{\rho_s}{\rho_{\rm crit}} = \frac{2}{d-1}$$

— unity only at $d=3$. This is the general-d formula cited in [SM §3.2] as the boundary-entropy filter selecting $d=3$.

This boundary-entropy filter is one of three independent $d=3$ selectors in the framework: (i) substratum coupling-degree minimization (SM Theorem 6, $K=2d$ gives $K=6$); (ii) Bravais-lattice uniqueness for the SM gauge group (SM Theorem 7b); (iii) the present boundary-entropy filter. The three derive $d=3$ from very different physical considerations (substrate dynamics, crystallographic representation theory, cosmological boundary thermodynamics). The convergence is structurally significant.

(ii) Invisibility. The argument has two logically distinct components.

(ii-a) Construction-level invisibility (from Part I alone). The boundary modes comprising $S_{\rm dS}$ are hidden-sector degrees of freedom — the variables traced out to produce the quantum description. By the trace-out structure, no hidden-sector degree of freedom maps to an operator on $\mathcal{H}_V$. This is a theorem of Part I, independent of any gravitational content: the hidden sector is invisible to the emergent description by construction.

(ii-b) Gravitational consequence (from the ontological ordering of §6.2). The physical import — that the boundary entropy's gravitational influence appears as additional gravity with no source in $\langle\hat{T}_{\mu\nu}\rangle$ — requires the geometry-first ordering established in §6.2: the metric exists at the pre-trace-out level, and the stress-energy sourcing gravity is the classical stress-energy, not the emergent quantum expectation value. Under this ordering, the gravitational interaction between bulk matter and boundary entropy is mediated by the metric at the classical level, modifying the geometry but contributing no term to the emergent $\langle\hat{T}_{\mu\nu}\rangle$. To the embedded observer, this appears as additional gravity with no particle-sector source.

The corollary's predictions are therefore conditional on the same geometry-first ordering that dissolves the CC problem in §6.2. This is not an additional assumption: the ordering is already required by Part II's logic (the partition is defined by null geodesics, so the metric must exist prior to the partition). What the corollary provides is a second, qualitatively distinct *consequence* of that ordering.

(iii) Budget. The emergent QFT accounts for the baryonic sector (~5% of ρ_{crit}). The remaining ~95% is boundary entropy — gravitationally active (because ρ_s is a classical quantity computed before the trace-out, unlike $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4$ which is an artifact of the emergent description) but invisible to the emergent QFT. This matches the observed composition, in which ~95% of the gravitational content has no source in particle physics. □

Observational status of particle dark matter searches. The corollary's prediction is that no particle dark matter exists — the dark sector is the trace-out, not a missing particle species. The strongest current constraint is the LZ Collaboration's 417-day analysis (December 2025) [21], which is the most sensitive WIMP search ever conducted and reports null results across the 3–9 GeV range, extending world-leading constraints from XENONnT and PandaX-4T to lower mass. The continued failure to detect particle dark matter across multiple decades of increasingly sensitive experiments — summarized in the literature as "the waning of the WIMP" — is consistent with the corollary's prediction. Definitive falsification of the corollary requires positive detection of a particle species accounting for the structured dark sector, which has not occurred.

Memory effects and P-indivisibility. Matter locally deforms the horizon geometry, redistributing the boundary entropy. This displacement persists because $\tau_B \gtrsim 1/H$ (condition C2): the hidden-sector modes carrying the redistributed entropy do not thermalize back to the unperturbed configuration on sub-Hubble timescales. Verlinde [20] identifies this as a "memory effect" from the non-thermalization of de Sitter states at sub-Hubble scales. In the present framework, this non-thermalization is not assumed but derived: it is P-indivisibility operating in the gravitational sector, with (C2) ensuring that entropy displacements persist without thermal overwriting.

The corollary is a consistency check for the total dark sector budget, not a derivation of its internal structure. The uniform component of the boundary entropy corresponds to dark energy (§6); the structured component — the matter-induced redistribution of boundary entropy producing local gravitational effects resembling dark matter — is derived in §7.3. The ratio $\rho_{\text{QFT}}/\rho_s \sim S_{\text{dS}} \sim 10^{122}$ confirms the parallel with §6.3: the information compression ratio and the gravitational occlusion fraction are two aspects of a single phenomenon.

Falsifiability. The corollary would be falsified by detection of dark matter particles accounting for the full ~27% structured dark sector, or by evidence that the dark sector's gravitational effects are unrelated to the boundary geometry.

7.3 Dark matter from entropy displacement

The total dark sector budget is established by §7.2. This section derives the structured component from the entropy displacement induced by baryonic matter.

Step 1: Entropy displacement (derived). Baryonic matter M_B is coupled to the boundary entropy through H_{VB} (condition C1). The boundary is a thermal reservoir at $T_{\mathrm{dS}} = \hbar H/(2\pi c k_B)$ (derived, §3.2). For a static matter configuration in gravitational equilibrium — a quasi-static process — the Clausius relation gives the entropy displacement:

$$\Delta S = \frac{M_B c^2}{k_B T_{\mathrm{dS}}} = \frac{2\pi M_B c^3}{\hbar H}$$

This follows from the generalized second law (Theorem A.6) applied to the matter-boundary system: the boundary's entropy reduction is Q/T for a reversible process, where $Q = M_B c^2$ and $T = T_{\mathrm{dS}}$. For dynamic configurations, the Clausius inequality $\Delta S \geq Q/T$ only strengthens the displacement.

Step 2: The critical acceleration (derived). The displaced entropy redistributes over the volume between the matter and the horizon. The Jacobson mechanism converts the entropy gradient into additional radial acceleration. The conversion is dimensional: in d spacetime dimensions, the factor relating the natural Hubble acceleration scale cH to the empirical MOND acceleration is $(d-3)/[(d-2)(d-1)]$. This derivation is due to Verlinde [20] (eq. 1.7), and follows from the interplay between volume-law entropy distributed over $(d-1)$ spatial dimensions and area-law entropy scaling with $(d-2)$ surface dimensions, together with the standard surface-density-to-acceleration relation via Gauss's law. In our universe ($d = 4$):

$$a_0 = \frac{(d-3)}{(d-2)(d-1)} \cdot cH = \frac{1}{2 \cdot 3} \cdot cH = \frac{cH}{6} \approx 1.2 \times 10^{-10}\ \mathrm{m/s}^2$$

This is the MOND critical acceleration — derived, not fitted. (Milgrom's empirical value: $a_0^{\mathrm{obs}} \approx 1.2 \times 10^{-10}$ m/s².) The factor 1/6 thus has dimensional origin: it is structurally fixed by $d = 4$ together with the volume-vs-area entropy interplay.

The OI framework's contribution beyond Verlinde's derivation is *grounding the volume-law entropy mechanism in derivable structure rather than postulate*. In Verlinde's framework, the de Sitter states' non-thermalization ("memory effect") is hypothesized; in OI, it is derived from condition C2 ($\tau_B \gg \tau_S$) operating in the gravitational sector. This addresses the consistency objection of [37] by replacing the postulated elasticity with a structural consequence of P-indivisibility at the cosmological boundary. The published Verlinde-MOND consistency debate is consequently relevant to the strength of the elasticity assumption in [20], not to the OI derivation, where the relevant content is structurally grounded.

Step 3: The deep-MOND regime. For baryonic acceleration $\ll a_0$, the displaced entropy dominates. The Jacobson mechanism converts the entropy gradient into curvature, giving $g_{\text{total}} = \sqrt{g_B \cdot a_0}$ and rotation velocity:

$$v^2 = r \cdot g_{\text{total}} = \sqrt{GM_B \cdot cH/6} = \text{constant} \qquad \Longrightarrow \qquad v^4 = GM_B \cdot cH/6$$

This is the baryonic Tully-Fisher relation. Flat rotation curves emerge with no dark matter particles and no free parameters beyond the baryonic mass.

Step 4: Interpolation constraints. The full interpolation between the Newtonian regime ($g_B \gg a_0$) and the deep-MOND regime ($g_B \ll a_0$) is constrained by the framework but not uniquely determined. The entropy displacement defines a potential Ψ_D with $g_D = -\nabla\Psi_D$, and the framework requires:

(i) *No free parameters.* g_D depends only on g_B and $a_0 = cH/6$ — no halo-specific parameters, no fitting constants.

(ii) *Conservative.* $\nabla \times g_D = 0$, since g_D derives from the entropy potential.

(iii) *Monotonic.* $\partial g_D/\partial g_B > 0$ — more baryonic mass produces more entropy displacement, hence more dark matter effect.

(iv) *Correct limits.* $g_D \to 0$ for $g_B \gg a_0$ (Newtonian); $g_D \to \sqrt{g_B \cdot a_0}$ for $g_B \ll a_0$ (MOND).

(v) *Analytic.* The entropy redistribution is a smooth function of the coupling structure — no discontinuities or phase transitions in the interpolation.

These constraints narrow the interpolation to a one-parameter family indexed by the transition steepness — a property of the particular bijection φ, in the same category as fermion masses (solution-specific rather than structural). The simplest member satisfying all five constraints:

$$g_D = \frac{g_B}{2}\left(\sqrt{1 + \frac{4a_0}{g_B}} - 1\right)$$

which is equivalent to $g_{\text{total}} = g_B \cdot \nu(g_B/a_0)$ with $\nu(y) = (1+\sqrt{1+4/y})/2$ — the "simple" MOND interpolation function corresponding to $\mu(x) = x/(1+x)$. This form matches the empirical radial acceleration relation (McGaugh et al. [22]) to < 0.1 dex across all galaxy-relevant accelerations.

Epistemic status. Steps 1–3 are Tier 2: the displacement formula follows from the Clausius relation applied to the boundary reservoir (§3.2, A.6), the geometric factor from spherical redistribution in de Sitter, and $a_0 = cH/6$ and $v^4 =$

$GM_B \cdot cH/6$ are parameter-free consequences. The interpolation constraints (i)–(v) are Tier 2; the transition steepness is Tier 3 (solution-specific).

Redshift evolution. Since $a_0(z) = cH(z)/6$ and $H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$, the critical acceleration increases with redshift: $a_0(z = 1) = 1.79\, a_0(0)$, $a_0(z = 2) = 3.03\, a_0(0)$, $a_0(z = 3) = 4.57\, a_0(0)$. The MOND crossover radius $r_M = \sqrt{6GM_B/(cH)}$ correspondingly shrinks — for a Milky-Way-like baryon budget ($M_B \approx 6 \times 10^{10}\, M_\odot$ at the stellar-plus-cold-gas level), $r_M = 9.0$ kpc at $z = 0$ but 5.2 kpc at $z = 2$ and 3.1 kpc at $z = 5$. Galaxies with effective radii comparable to or exceeding r_M show flat rotation curves (the familiar local behavior); galaxies with $r_{\text{eff}} < r_M$ show declining, Keplerian-like rotation curves. Since r_M shrinks with increasing z, high-redshift galaxies are systematically more baryon-dominated — their rotation curves decline at smaller radii.

This prediction is borne out by the data. Genzel et al. [23, 24] report stacked rotation curves for massive star-forming galaxies at $z = 0.9$–2.4 showing a declining outer velocity at $> 3\sigma$ significance relative to local spirals — consistent with the OI prediction. In ΛCDM, flat rotation curves from NFW halos are expected at all redshifts; explaining the Genzel data requires anomalously low halo concentrations or baryon-dominated inner regions. The cleanest local test is Jiao et al. [25], who detect for the first time a Keplerian decline in the Milky Way's own rotation curve from ~ 19 to ~ 26.5 kpc using Gaia DR3 kinematics, with the flat rotation curve hypothesis rejected at 3σ. Our own galaxy is the strongest single piece of evidence for the $H(z)$-dependent crossover prediction at $z = 0$, where the OI framework predicts the crossover at $r_M \sim 17$ kpc for the Milky Way's full baryon budget including the hot circumgalactic gas halo ($M_B \sim 2 \times 10^{11}\, M_\odot$) — essentially where Jiao et al. observe the transition.

Tully-Fisher tension and resolution. McGaugh et al. (2024), analyzing Nestor Shachar et al. (2023) JWST kinematics, report no evolution in the Tully-Fisher relation out to $z \sim 2.5$, seemingly contradicting $v^4 \propto H(z)$. However, this comparison uses *stellar* mass M_*, not baryonic mass $M_B = M_*/(1 - f_{\text{gas}})$. At high redshift, gas fractions are large ($f_{\text{gas}} \sim 50$–70% at $z \sim 2$; Tacconi et al. 2018). The OI-predicted shift in the baryonic TF ($\Delta \log M_B = \log(H_0/H(z)) = -0.48$ dex at $z = 2$) is almost exactly compensated by the gas mass not included in M_*: the cancellation requires $f_{\text{gas}} = 1 - H_0/H(z)$, which gives 44% at $z = 1$, 58% at $z = 1.5$, and 67% at $z = 2$ — squarely in the observed range. The apparent "no evolution" in the stellar TF is therefore *predicted* by the OI framework: the dynamical shift (higher a_0) and the compositional shift (higher gas fraction) cancel when only stellar mass is measured. The definitive test is the *baryonic* TF at $z > 1$ with reliable ALMA gas masses: the OI framework predicts a normalization shift of -0.48 dex at $z = 2$ at fixed v_{flat}.

Observational predictions. Three parameter-free, quantitative tests:

(i) *Rotation curve evolution.* The outer rotation curve evolves with z through

$a_0(z) = cH(z)/6$. For a $10^{10.8}\,M_\odot$ exponential disk with $r_{\rm eff} = 5$ kpc, the outer slope $\Delta v/v$ between $r_{\rm turn}$ and $2r_{\rm turn}$ is -0.08 at $z = 0$, evolving to -0.06 at $z = 2$ and -0.05 at $z = 5$ — flattening as the disk increasingly sits in the deep-MOND regime, while the asymptotic $v_{\rm flat}(z) = (GM_B \cdot cH(z)/6)^{1/4}$ shifts upward by $E(z)^{1/4}$ (+32% at $z = 2$). The Genzel et al. declining $z = 1$–2 rotation curves are consistent with this picture: their observed declines reflect rotation curves entering MOND-flat regimes at higher $v_{\rm flat}$ than at $z = 0$, not steeper Keplerian declines. Particle dark matter predicts no evolution in either the outer slope or in $v_{\rm flat}$ at fixed M_B.

(ii) *Tully-Fisher evolution.* The baryonic Tully-Fisher relation evolves: $v_{\rm flat}^4 = GM_B \cdot cH(z)/6$. At fixed baryonic mass, $v_{\rm flat} \propto H(z)^{1/4}$: 7% higher at $z = 0.5$, 16% higher at $z = 1$, 32% higher at $z = 2$. Equivalently, at fixed $v_{\rm flat}$, the inferred baryonic mass at $z = 2$ is $H_0/H(z = 2) = 0.33\times$ the local value — galaxies appear to need less dark matter at high z. Testable with JWST and ALMA kinematics combined with photometric baryonic mass estimates.

(iii) *No halo-specific parameters.* In particle DM, each galaxy has a dark matter halo characterized by at least two free parameters (concentration and mass). In the OI framework, the dark matter effect at radius r depends only on the enclosed baryonic mass $M_B(< r)$ and $H(z)$ — no halo-specific parameters. This predicts: the radial acceleration relation (RAR) at fixed g_B has zero *intrinsic* scatter (at fixed z) — observational scatter is set by measurement uncertainties — and the *intrinsic* scatter does not correlate with halo properties. Current SPARC data [22] show remarkably small RAR scatter (~ 0.13 dex), consistent with OI but requiring fine-tuning in ΛCDM.

Falsifiability. The entropy displacement account would be falsified by: (a) detection of dark matter particles accounting for $ $27% of $\rho_{\rm crit}$; (b) high-redshift rotation curves inconsistent with the predicted $H(z)$ dependence — specifically, if galaxies at $z > 2$ show flat rotation curves at $r > r_M(z)$; (c) baryonic Tully-Fisher violations exceeding baryonic mass uncertainties; (d) RAR scatter correlating with halo properties at fixed g_B and z.

Cluster-scale status. Galaxy clusters present the most challenging regime for MOND-like theories. For the Coma cluster ($M_B \approx 1.4 \times 10^{14}\,M_\odot$, $R_{200} \approx 1.9$ Mpc), the acceleration at the virial radius is $g_B/a_0 \approx 0.05$ — deep in the MOND regime. With the simple interpolation function above, the predicted velocity is $v = g_B \cdot \nu(g_B/a_0) \cdot R_{200} \approx 1260$ km/s; the observed velocity dispersion implies ~ 1270 km/s — a match to better than 1%. For other rich clusters (Perseus, A2029, A1689), the simple interpolation reduces the standard MOND mass shortfall from a factor ~ 2 to ~ 1.0–1.5; the residual is **not** closable by undetected cluster baryons. Modern censuses find massive clusters ($M_{500} \gtrsim 2 \times 10^{14}\,M_\odot$) baryon-complete to within $\sim 7\%$ of the cosmic fraction inside r_{500} (Planck cosmology), and consistent with the cosmic value at the virial radius;

the late-time missing baryons localized by FRB dispersion measures (Connor et al. 2025) reside in the diffuse IGM and cosmic filaments — *between* clusters, not inside them. The earlier appeal to WHIM adding 30–100% conflated the *cosmic* missing-baryon fraction (filamentary) with a cluster-internal reservoir that the data do not support. Nor is the residual absorbable by massive neutrinos: the framework predicts $\Sigma m_\nu \approx 0.059$ eV (§ above), two orders of magnitude below the ~ 2 eV that would be required. The rich-cluster residual is therefore an **open problem**. It most plausibly reflects the unresolved transition between the *local* MOND-like interpolation used here (galaxies, cluster dynamics) and the *CDM-like* linear behavior of the same entropy displacement established for the CMB and Bullet Cluster: a single field cannot be MOND-like in cluster cores and CDM-like in linear growth without a specified crossover, and deriving that crossover from the non-local relaxation kernel is deferred to the substratum-level analysis (the same uncomputed kernel as §7.3). The deep-MOND limit $v^4 = GM_B \cdot a_0$ (Tier 2) is identical for all interpolation functions; only the intermediate-acceleration behavior (Tier 3) differs. Galaxy-scale predictions are unaffected: at $g_B/a_0 = 0.5$ (typical outer disk), different interpolation functions agree to < 0.07 dex — well within the RAR scatter of 0.13 dex.

Colliding clusters (Bullet Cluster). The Bullet Cluster (1E 0657-56) presents the most-cited evidence for particle dark matter: after the collision, gravitational lensing peaks coincide with the galaxy concentrations ($\sim 15\%$ of baryonic mass), not with the X-ray gas ($\sim 85\%$ of baryonic mass) that was ram-pressure stripped to the center. In a local MOND picture where g_D tracks the instantaneous baryonic distribution, the lensing should peak where the gas is — contradicting observations. The OI framework resolves this through the non-local character of entropy displacement. The dark gravity arises from boundary entropy redistribution at the cosmological horizon, and the horizon's response time is $\tau_{\text{relax}} \sim H^{-1} \approx 14$ Gyr. The relaxation timescale identification ($\tau_{\text{relax}} \sim 1/H$) is itself a Tier 2 derived quantity from the de Sitter boundary's response to perturbations; specific calculations of mode-by-mode relaxation are deferred to a substratum-level analysis. The collision crossing time is $t_{\text{cross}} \sim 720\,\text{kpc}/4700\,\text{km/s} \approx 0.15$ Gyr — only 1% of the relaxation time. The entropy displacement is therefore *frozen* at the pre-collision configuration, centered on the galaxy concentrations (which defined the gravitational potential wells for gigayears before the collision). The gas moved during the collision; the boundary entropy did not. This reproduces the observed morphology: lensing peaks at the galaxy positions, gas peak between them — without invoking collisionless particles. The mechanism makes a testable prediction: for very old post-collision systems ($t_{\text{since}} \gg 1$ Gyr), the dark gravity should gradually relax toward the gas distribution on timescale $\sim H^{-1}$. Particle dark matter predicts no such relaxation.

CMB acoustic peaks. The CMB power spectrum provides the most precise test of the dark matter paradigm: the odd/even peak height asymmetry requires a non-oscillating gravitational source (CDM in ΛCDM) that provides potential wells for the photon-baryon fluid to oscillate in. The entropy displacement satisfies this requirement through the same thermodynamic averaging that resolves the Bullet Cluster. The Clausius relation $\Delta S = \delta Q/T_H$ involves the *net* heat transfer to the horizon boundary. For an oscillating perturbation (compression then rarefaction), the net entropy displacement per acoustic cycle is zero — the oscillating component averages out. Only the secular (growing) component produces net displacement. The entropy displacement perturbation therefore does not oscillate with the photon-baryon fluid; it tracks the growing mode exclusively.

The entropy displacement perturbation then evolves identically to CDM in the linear regime: (i) it has the same effective density ($\rho_E \approx \Omega_{\rm CDM} \times \rho_{\rm crit}$, from the dark-sector corollary §7.2); (ii) it does not couple to photons (it is a gravitational effect at the horizon, not a fluid component); (iii) it grows under its own gravity from adiabatic initial conditions set by inflation; (iv) its growth equation $\ddot{\delta}_E + 2H\dot{\delta}_E = 4\pi G\rho_{\rm eff}\delta_E$ has the same source term as CDM. The CMB power spectrum — which depends on Ω_b/Ω_m (peak ratios), the growth of perturbations before recombination (peak heights), the sound horizon r_s (peak positions), and the damping scale (high-ℓ suppression) — is therefore predicted to match ΛCDM at linear order. A direct CLASS comparison (2026-04-26) confirms this: with the framework's Type II RVM at $\nu = 10^{-4}$ (the Bertini 2025 1σ high end) encoded via CLASS's CLP fluid dark-energy module, the four acoustic peak amplitudes (TT) match ΛCDM to $\lesssim 0.02\%$, full-range D_ℓ^{TT} residuals have median 0.006% and maximum 0.08%, and corresponding D_ℓ^{EE} residuals are similar — all 20–300× below cosmic variance and 5–50× below Planck 2018 binned precision at every ℓ from 2 to 2500. *Epistemic status: Tier 2. The CLASS run verifies the residual RVM dark-**energy** effect on the spectrum (the dark matter is modeled as standard CDM, the dark-energy sector as the CLP-fluid RVM); it does not by itself test the displacement ≡ CDM step, which is the analytic result (i)–(iv) above. The temporal half of that equivalence — that a single relaxation kernel reproduces CDM-like growth at recombination while keeping the Bullet frozen — is demonstrated below; a direct Boltzmann run implementing the displacement kernel explicitly, including its k-dependence, is the remaining numerical check.*

Relaxation kernel: Bullet–CMB consistency. The frozen-displacement account of the Bullet Cluster (slow $\tau_{\rm relax}$) and the growing-mode tracking the CMB requires (fast response at recombination) are reconciled by a single kernel once $\tau_{\rm relax}$ is read as the *epoch-local* horizon time $\tau_{\rm relax}(z) \sim 1/H(z)$ — the ≈ 14 Gyr quoted above is its $z \approx 0$ value, not a fixed constant. Because the linear growing mode also evolves on $\sim 1/H(z)$, the displacement relaxation equation $d\delta_E/dt = (\delta_m - \delta_E)/\tau_{\rm relax}$ becomes, in scale-factor form, $d\delta_E/da = (\delta_m - \delta_E)/a$ — *epoch-independent*, with solution $\delta_E = \frac{1}{2}\,\delta_m + \mathcal{O}(a_{\rm rec}^2/a)$. Thus $\delta_E \propto a$ tracks the CDM growing mode at every epoch with no lag at recombination and no temporal cutoff; the constant ratio is absorbed into the calibration $\rho_E \approx \Omega_{\rm CDM}\rho_{\rm crit}$ (§7.2). A *fixed* $\tau_{\rm relax} = 1/H_0$ would instead suppress the response by $H_0/H(z) \sim 1/10$ at $z = 100$ and distort the peaks — confirming that the epoch-local reading is the correct one (numerically verified: δ_E/δ_m constant to $< 0.1\%$ across $z = 1100 \to 0$ for $\tau = 1/H(z)$, versus an order-of-magnitude lag for $\tau = 1/H_0$). The Bullet remains frozen for the independent reason that a cluster collision is anomalously fast on the cosmic clock: $t_{\rm cross}/\tau_{\rm relax}(z\!=\!0.3) \approx 0.15/12 \approx 0.013 \ll 1$. The two regimes are therefore controlled by *different* ratios — $\tau_{\rm relax}/t_{\rm growth} \sim \mathcal{O}(1)$ at all epochs (tracking) versus $\tau_{\rm relax}/t_{\rm collision} \gg 1$ (freezing) — and do not conflict. Since the Jacobson sourcing is spatially local, the non-locality is *temporal* (a memory over $\tau_{\rm relax}$), not spatial, and imposes no k-cutoff on the linear growth. *Open computation (not an inconsistency):* the k-dependence of the relaxation rate, required for a full mode-by-mode kernel. *Epistemic status: Tier 2 (temporal reconciliation verified numerically, relaxation_kernel_check.py 2026-06-29; k-dependence deferred).*

7.4 Summary of GR-side predictions

The GR-side quantitative predictions of the framework are collected in the table below. Each follows from partition geometry at the cosmological horizon; no predictions are fit to the quantities they predict.

Prediction	Formula / value	Observed / status	Match
Emergent action scale	$\hbar = c^3(2l_p)^2/(4G)$	$\hbar_{\rm obs} = 1.055 \times 10^{-34}$ J $\cdot$ s	structural
Discreteness scale	$\epsilon = 2\,l_p$	Planck length	structural
BH entropy coefficient	$S_{BH} = A/(4l_p^2)$	derived ($2\pi/8\pi$); no direct test — GW250114 confirms the classical area theorem, which the framework preserves	consistent (non-discriminating)
Cosmological constant	$\rho_\Lambda \sim H^2/G$	observed value	no discrepancy
Dark energy form (structural)	Type II RVM (functional form derived; magnitude $\approx \lvert\nu_{\rm QFT}\rvert \sim 10^{-32}$, indistinguishable from zero)	Bertini 2025 $\nu = -(2.5{\pm}1.3){\times}10^{-4}$	consistent at 2σ (Bertini consistent with zero)
Dark sector fraction	$\Omega_{\rm dark} \approx 95\%$	$\Omega_{\rm CDM} + \Omega_\Lambda \approx 95\%$	consistent
MOND acceleration scale	$a_0 = cH/6 \approx 1.2 \times 10^{-10}$ m/s²	$a_0^{\rm obs} \approx 1.2 \times 10^{-10}$ m/s²	exact
Baryonic Tully-Fisher	$v^4 = GM_B \cdot cH/6$	SPARC data	within 0.13 dex RAR scatter

Prediction	Formula / value	Observed / status	Match
Rotation curve crossover	$r_M = \sqrt{6GM_B/(cH)}$	Milky Way: ~ 17 kpc	Jiao et al. 2023, 3σ
High-z rotation curves	More baryon-dominated with z	Genzel et al. $z = 0.9$–2.4	consistent at $> 3\sigma$

The Type II running-vacuum form for dark energy is the gravitational sector's most distinctive *structural* prediction: the functional form is forced by boundary-entropy architecture (§7.1), the Planck-scale UV cutoff is forced by the gap equation (§3.2, §4), and matter conservation plus Bianchi consistency fixes the coupled running of $G(H)$ and $\rho_{\text{vac}}(H)$. The *magnitude* of ν is given at theorem level by the self-consistent emergent-QFT calculation: $|\nu_{\text{QFT}}| \sim (M_{\text{SM}}/M_{\text{Pl}})^2 \sim 10^{-32}$, indistinguishable from zero. Substratum contributions beyond this are constrained by the boundary mode count of §3.1 and §5, which catalogues modes by spatial location with no inherent frequency assignment; the natural 2D field-theoretic distribution concentrates modes near the UV cutoff and contributes $|\nu|_{\text{substratum}}$ well below $|\nu_{\text{QFT}}|$. The framework's theorem-level prediction is therefore $|\nu| \approx |\nu_{\text{QFT}}|$, observationally indistinguishable from zero. The Bertini et al. (2025) measurement $\nu = -(2.5 \pm 1.3) \times 10^{-4}$ is consistent with this prediction at 2σ, where Bertini is itself consistent with zero. DESI Year 5 ($\sim 1\%$ precision on ν) provides the decisive empirical test: a null confirms the prediction; a high-confidence detection at the 10^{-4} level would require additional substratum structure (specifically, log-uniform mode-per-decade distribution at the cosmological horizon) not currently derivable from §3-§5. The functional form (Type II RVM) is structurally protected and is the framework's pre-registered prediction.

Remark (the magnitude question reduces to emergent locality). The choice between the two boundary-mode spectral measures is not an independent free parameter: it is the same question as the radiative emergent-Lorentz problem ([SM §3.1, point iii]). The natural 2D field-theoretic measure ($dN/d\omega \propto \omega$, giving $|\nu| \approx |\nu_{\text{QFT}}|$) is the one induced when the coarse-grained boundary dynamics inherits the locality of the range-1 substratum; the log-uniform measure ($dN/d\ln\omega = \text{const}$, giving $|\nu| \sim 10^{-4}$) is not the spectrum of a local field theory in any dimension, and would require the coarse-graining to generate scale-invariant (non-local) boundary dynamics — precisely the failure mode that would also generate a relevant Lorentz-violating operator. The two open questions therefore share one premise: whether the substratum-to-observer coarse-graining preserves locality. A consequence is that the DESI Year 5 measurement is effectively a test of emergent horizon locality, with a built-in cross-check: a detection of $|\nu|$ at the 10^{-4} level would, under this analysis, require emergent non-locality, which must independently manifest as Lorentz violation — so a positive ν cannot be absorbed as a free parameter without a correlated Lorentz-violation signature. This reduces the framework's count of independent open questions; the magnitude of ν is forced to ≈ 0 conditional on the same emergent locality the framework requires for Lorentz invariance, rather than being independently open. (The reduction is an argument at the level of spectral-measure phase-space counting plus locality inheritance, not yet a lattice computation of the boundary spectral measure; it is presented as a structural linkage, with the underlying coarse-graining-preserves-locality step remaining the open item shared with [SM §3.1].)

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8. Discussion

The foundational emergent QM result is treated as established in [Main] and not re-derived here. The cubic-lattice realization of the substratum — which determines the Standard Model gauge structure via the wave equation — is independent of the present results and is developed in [SM]. The reconstruction theorem, the substratum gauge group, and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S, φ) are developed in [Substratum].

8.1 Scope

The characterization theorem is a full equivalence: QM $\iff$ embedded observation under (C1)–(C3), conditional on the hidden sector satisfying the eigenstate thermalization hypothesis (ETH) for the C2 direction (see [Main, §3.4]). The C1 and C3 necessity directions are unconditional. For the cosmological horizon application, ETH holds for any generic chaotic hidden sector — the horizon complement is such a system — so the full equivalence applies in our universe. The theorem does not identify which physical systems satisfy the conditions (empirical) nor derive $\hbar$ for specific systems (requires Part II). Universality in our universe follows from the shared causal partition defined by null geodesics.

8.2 The status of the discreteness scale

ϵ does not smuggle in a quantum assumption. Lemma 1 requires finite-dimensionality, motivated by the classical fact that finite-area boundaries admit finitely many modes. The result $\epsilon = 2\,l_p$ is a *consequence* of self-consistency (§4). Holographic entropy bounds [26] provide independent, non-quantum motivation.

8.3 Vacuum energy: relative effects and the Higgs potential

Relative effects. Casimir forces and Lamb shifts are predictions *within* the emergent QFT — relative effects depending on energy differences between configurations. The framework denies that the *absolute* zero-point sum gravitates, because gravity operates at the logically prior classical level (§6.2).

The Higgs potential. The strongest CC objection is not the zero-point sum but the electroweak Higgs potential: symmetry breaking shifts $V(\phi)$ by $\sim$ (200 GeV)^4, exceeding the observed Λ by $\sim$ 55 orders of magnitude. The framework's response follows from its ontological ordering and the structural/volumetric distinction of §3.2, applied as a precise principle:

What gravitates: Relative energy differences between configurations of the emergent fields. An electron at rest vs. no electron differs by $m_e c^2$ in the classical substratum — there is a classical microstate difference corresponding to this energy difference. Relative differences appear in the classical stress-energy tensor and gravitate. Casimir forces, Lamb shifts, and particle masses all fall in this category.

What does not gravitate: Absolute energy levels assigned to configurations by the emergent description. The Higgs vacuum energy shift $V(\phi_{\min})$ is the energy the emergent QFT assigns to its ground state — the configuration obtained after symmetry breaking. But the "ground state" in the emergent description corresponds to a class of classical microstates whose energy in the classical substratum is $\rho \sim H^2/G$, not $\rho \sim$ (200 GeV)^4. The shift is a property of how the emergent description organizes the microstates (the correspondence), not a property of the microstates themselves (the classical dynamics).

The principle is uniform: the zero-point sum, the Higgs vacuum energy, and the QCD condensate energy are all absolute properties of the emergent ground state. The dissolution applies to each for the same reason — gravity operates at the classical level (§6.2), where none of these quantities appear.

Why QCD binding energy gravitates but the quantum vacuum does not. The proton mass is ~99% QCD binding energy — a purely quantum effect. If gravity couples to the classical substratum, how does it "know" about masses generated by emergent quantum dynamics? The answer is that QCD binding energy and vacuum zero-point energy occupy different categories in the structural/volumetric distinction of §3.2. Relative energies within the emergent QFT — energy differences between configurations — correspond to differences in the coupling structure of φ, which the Jacobson derivation reads as curvature. A proton vs. three free quarks corresponds to two distinct classes of classical microstates with different energies in the classical substratum; the binding energy $\sim$ 938 MeV is a *relative* quantity that appears in H_{tot} and gravitates. The absolute baseline — the zero-point energy summed over all modes — corresponds to the total information capacity of the hidden sector (S_{dS}), which is a property of the partition geometry, not of the dynamics, and therefore does not source curvature.

8.4 Assumptions, limitations, and falsifiability

The theorem requires Lemma 1 (independent in Part I; consequence of geometry in Part II), the stochastic-quantum bridge ([Main, §3.1] and Appendix A; established by two independent routes), and genericity conditions (non-degenerate spectrum, non-vanishing overlaps), which hold for all but a measure-zero set. The P-indivisibility proof uses finiteness via the recurrence argument ($\varphi^N = \mathrm{id}$). The deep-sector cardinality corollary (§3.2) establishes that only $\mathcal{C}_V$ and $\mathcal{C}_B$ need be finite — both guaranteed by holographic entropy bounds [26] independently of the de Sitter finite-dimensionality conjecture — while the deep sector $\mathcal{C}_D$ may be infinite. The accessible-timescale backflow lemma ([Main, §2.3]) provides a second route to P-indivisibility that is independent of $|\mathcal{C}_H|$. The characterization theorem becomes an equivalence on observable timescales ($T_{\text{obs}} \ll \tau_B \sim 10^{17}$ s) — physically indistinguishable from a timeless identity for all experiments within the age of the universe.

The sharp partition and constraint structure. Lemma 2's product decomposition $\Gamma = \Gamma_V \times \Gamma_H$ is the kinematic phase space. In GR, the Hamiltonian and momentum constraints restrict the physical phase space to a submanifold $\Gamma_{\text{phys}} \subset \Gamma_V \times \Gamma_H$ on which interior and exterior data are correlated. P-indivisibility survives on a constraint surface because restoring P-divisibility would require decoupling the sectors entirely, violating the constraints; the constraint surface thus enforces (C1) non-perturbatively. For systems other than the cosmological horizon, the fidelity of the product approximation determines the fidelity of the emergent quantum description.

Planck-scale ordering. The classical-prior ontology may break down at l_p, where "geometric" and "quantum" lose operational distinction. If a deeper theory unifies both, this framework describes the effective macroscopic regime.

Falsifiability. The *theorem* would be falsified by a failure of both the stochastic-quantum correspondence and the Stinespring construction for the class of processes generated here — a possibility excluded by the independent proofs in [Main, §3.1] and Appendix A. The *cosmological application* (Type II RVM functional form as a structural prediction) would be falsified by definitive observational demonstration that dark energy deviates from the ν-parameterized Type II form — e.g., a high-confidence detection of strong $w(z)$ variation incompatible with $w = -1 + \mathcal{O}(\nu)$. The theorem-level magnitude prediction $|\nu| \approx |\nu_{\text{QFT}}| \sim 10^{-32}$ admits a stratified falsification structure: a measurement $|\nu| \gtrsim 10^{-4}$ at high significance would require log-uniform boundary-mode spectral structure beyond §3-§5; a measurement $|\nu| > 10^{-3}$ at high significance would constitute a structural challenge to the magnitude prediction without falsifying the Type II functional form itself. The *dark-sector concordance* (§7.2) would be falsified by detection of dark matter particles accounting for the full structured dark sector, or by evidence that the dark sector's gravitational effects are unrelated to the cosmological boundary geometry.

The falsifiability content is stratified by hierarchy level (per [Structure §2.4]). The Type II RVM functional form is a Tier 1 / Level C $\times$ Level G4 prediction — its falsification would refute the universality-class structural-realism claim that any horizon-bounded embedded-observer system produces this form, not specifically OI's cubic-lattice substratum. The dark-sector concordance is a Tier 2 / Level D $\times$ Levels G1–G2 prediction — falsification would refute OI's specific universality-class representative without necessarily refuting the broader Tier 1 framing. The functional-form vs. magnitude prediction (e.g., $|\nu|$ at 10^{-3} vs. $|\nu| \sim 10^{-32}$) tracks this stratification: functional-form falsification operates at the universality-class level, magnitude falsification at the specific-representative level.

A near-term mechanism-side test. A more tractable falsification target is provided by the recently developed operational distinction between classical and quantum memory in non-Markovian processes [27, 28]. Some non-Markovian quantum dynamics can be simulated using only classical stored information, while others require genuinely quantum memory in the environment, with the dividing line set by entanglement structure of the channel's Choi state. The framework predicts that all *fundamental* non-Markovian dynamics in nature — those arising from the cosmological trace-out rather than from engineered quantum baths — should be classical-memory simulable, because the hidden sublattice is by construction a classical substrate. A physically realized fundamental open-system process whose memory provably exceeds the classical bound would falsify the framework's foundational ontology even without cosmological data. This is a near-term tractable test in a literature where the experimental and theoretical tools are advancing rapidly.

Black-hole horizons under partition-relativity. The framework's mechanism is grounded in the cosmological horizon as the primary partition for any sub-c observer in this universe (§2.1). A black-hole horizon within the cosmological visible sector is a *local* causal boundary: it prevents an external observer from probing the black-hole interior by direct geometric access, but it does not redefine that observer's epistemic partition in the sense of [Main, §1.4]. Matter falling into a distant black hole was already inside the external observer's Γ_V before the merger and remains inside it after; the trace-out that defines the external observer's emergent quantum description is over the cosmological horizon, not over the black-hole horizon. The framework therefore predicts that gravitational wave signatures from binary black-hole mergers follow standard general-relativistic ringdown morphology, with no OI-specific echo, comb, or wall-reflection features. A four-event coherent matched-filter stack on LVK O1–O4 strain data using a dedicated discreteness-scale echo template (with the predicted detector-frame delay $\Delta t = (r_+/c)\ln(r_+/2l_p)$ at the remnant mass and spin) returned a clean null: stacked statistic -0.45, empirical $p = 0.66$, with the previously hypothesized echo amplitude excluded at 5.9σ. The framework's BH physics for external observers is identical to standard GR plus the cosmological-scale dark-sector effects of §7.2–§7.3.

Nested and time-dependent partitions. The primary partition for any sub-c observer in this universe is the cosmological horizon (§2.1). When additional causal boundaries are present — most naturally, a black hole horizon within the cosmological patch — the trace-out machinery extends to a nested formalism, addressed in Appendix A. The nested formalism is a self-consistency exercise: it shows that the framework's emergent description is well-defined when multiple boundaries coexist, and that quantities like the generalized second law and the Page curve emerge naturally. It is *not* a claim that external observers' emergent quantum descriptions acquire BH-horizon-specific features; under partition-relativity (above), the cosmological horizon remains the primary partition. Appendix A resolves the formal nested-partition machinery in three layers.

(i) Consistency of nested trace-outs (Appendix A, Theorems A.1–A.4). Two procedures — tracing out the trans-cosmological region and the black hole interior simultaneously (direct) or sequentially (first cosmological, then black hole) — produce the same CPTP channel, the same emergent Hamiltonian (up to D-gauge), and the same $\hbar$. The key result (Theorem A.2) is that the quantum partial trace within the emergent description agrees with the classical marginalization on the substratum: the emergent QM is self-consistent.

(ii) The generalized second law (Appendix A, Theorem A.6). The sum $S_{\mathrm{matter}} + S_{\mathrm{BH}} + S_{\mathrm{dS}}$ is non-decreasing, proved from strong subadditivity and CPTP monotonicity of the emergent quantum description. Any process that shrinks the black hole (decreasing S_{BH}) transfers information to the visible sector and cosmological boundary, compensating the decrease.

(iii) The Page curve (Appendix A, Theorems A.7–A.9). For an evaporating black hole, the entanglement entropy of the radiation follows $S_{\mathrm{rad}}(t) = \min(n_R(t), n_B(t)) \cdot \ln q$ plus exponentially small corrections, where $n_B(t)$ is the shrinking boundary mode count and $n_R(t) = n_B(0) - n_B(t)$. Unitarity is preserved (bijectivity of φ), the turnover occurs at the Page time $t_P \approx 0.646\, t_{\mathrm{evap}}$ (structural, independent of mass and greybody factors), and the Page value is recovered for all but an exponentially negligible fraction ($\sim e^{-10^{77}}$ for $30 M_\odot$) of initial conditions — proved from bijectivity (cycle decomposition on the finite energy shell) and Popescu-Short-Winter typicality [29], without assuming ergodicity. The greybody factor is solution-specific.

Laboratory tests of the characterization theorem may be possible in analogue gravity systems where the hidden sector capacity is tunable.

8.5 Universality-class character of Tier 1 results

§1 distinguished Tier 1 results — the derivation of $\hbar$, the discreteness scale $\epsilon = 2\, l_p$, the Bekenstein-Hawking area law with the 1/4 coefficient, the dissolution of the cosmological constant problem, and the Type II running-vacuum functional form of dark energy — as following from partition geometry alone, not from specific substratum dynamics. This section makes that universality claim explicit and identifies what structural conditions any candidate substratum must satisfy for the Tier 1 results to apply.

The universality claim. Let $\mathcal{S}$ be any finite deterministic substratum (not necessarily OI's cubic lattice) admitting a partition into a visible sector V and hidden sector H at a cosmological-horizon-like causal boundary, satisfying [Main]'s structural conditions C1 (non-trivial coupling), C2 (slow bath: $\tau_S \ll \tau_B$), and C3 (high capacity: $N_H \gg N_V$). Then the following Tier 1 results follow from partition geometry and thermal self-consistency alone:

(T1.1) *Emergent action scale.* $\hbar = c^3\epsilon^2/(4G)$, where ϵ is the partition's discreteness scale.

(T1.2) *Discreteness scale.* $\epsilon = 2\,l_p$ exactly, fixed by the simultaneous solution of the action-scale relation and the definition of the Planck length.

(T1.3) *Area law.* The boundary entropy is $S = A/(4\,l_p^2)$, with the 1/4 coefficient fixed as the exact ratio $2\pi/8\pi$ between the Euclidean KMS period and Jacobson's Einstein-equation prefactor.

(T1.4) *Cosmological constant dissolution.* The emergent quantum vacuum energy ($\sim M_{\rm Pl}^4$) and the effective cosmological constant ($\sim H^2/G$) refer to logically distinct objects across two levels of description; the 10^{122} ratio is a compression ratio, not a failure of cancellation.

(T1.5) *Type II running-vacuum functional form.* Dark energy density obeys $\rho_\Lambda(H) = \rho_\Lambda^0 + \nu(H^2 - H_0^2)M_{\rm Pl}^2/(8\pi)$ with ν controlled by the boundary mode count, with theorem-level magnitude $|\nu| \approx |\nu_{\rm QFT}| \sim (M_{\rm SM}/M_{\rm Pl})^2 \sim 10^{-32}$.

What the derivations actually depend on. Reviewing §3–§7:

- The $\hbar$ *derivation* (§3) uses: the classical horizon temperature from the Jacobson thermodynamic argument (depends only on classical horizon geometry); the Euclidean KMS period of the emergent QFT on the horizon (depends only on the partial-trace structure of the channel and the horizon's discreteness scale); thermal self-consistency requiring agreement between the two (depends on the consistency of emergent QFT with classical thermodynamics at the horizon). None of these depends on OI's specific cubic-lattice structure; they depend on the existence of a cosmological-horizon partition, the C1-C3 conditions, and the Stinespring/partial-trace machinery [Main §3]. One further dependency is carried by the KMS step of Step 4 and should be stated: the Euclidean-regularity argument presupposes the emergent theory has local boost structure at the horizon (a bifurcate Killing horizon with a well-defined imaginary-time circle). That prerequisite is the local-boost-invariance of the emergent description — the same object whose protection is the framework's open Lorentz-sector item in the boost channel ([Main §3.5]; the custodial normalization and M_X window, [Substratum §6.4]). The $\hbar$ derivation is therefore not parallel to the Lorentz-violation accounting but partially downstream of its boost sector: a failure of emergent local boost invariance at the horizon would propagate to the gap equation, whereas the rotational sector does not enter here (spatial anisotropy does not touch the imaginary-time periodicity). The UV-dispersion robustness handled above ($\mathcal{O}((\epsilon\kappa/c)^2)$ corrections) is a separate and weaker condition; the boost-structure prerequisite is the load-bearing one and is inherited, not established, in this paper.

- The *discreteness scale fixing* (§4) is purely algebraic: rearranging the $\hbar$ relation gives $\epsilon^2 = 4\hbar G/c^3 = 4l_p^2$. Independent of substratum dynamics.
- The *area law derivation* (§5) uses: the boundary mode count $N_{\text{modes}} = A/\epsilon^2$ (depends only on the boundary's geometric area and the discreteness scale); the identification of mode count with entropy via the partial-trace structure. Independent of substratum dynamics beyond the C1-C3 conditions.
- The *CC dissolution* (§6) uses: the structural distinction between the emergent quantum vacuum energy (a property of the emergent QFT description) and the effective cosmological constant (a property of the classical substratum's gravitational dynamics). The distinction is structural — it follows from the two-level description architecture of any embedded-observer system, not from OI's specific substratum.
- The *Type II RVM functional form* (§7.1) uses: boundary entropy as bookkeeping degree of freedom in the gap equation; the Clausius relation $\delta Q = T\delta S$ at the horizon; the fact that boundary entropy is finite and depends on H via $A \propto H^{-2}$. The functional form $\rho_\Lambda \propto H^2$ follows from the gap equation's structure, which depends on the cosmological-horizon geometry but not on OI's specific substratum.

Sufficient structural conditions. The Tier 1 results require the following structural conditions on the substratum:

(*S1*) *Cosmological-horizon partition.* The substratum admits a partition V at a causal horizon with timescale $\tau_B \sim H^{-1}$ and information capacity $S_{\text{dS}} \sim 10^{122}$.

(*S2*) *C1–C3 satisfied.* The partition has substantial coupling (C1), slow-bath structure (C2), and high capacity (C3), per [Main §3].

(*S3*) *Finite discreteness.* The substratum has a finite discreteness scale ϵ that admits identification with the boundary's mode-counting unit. [Main §3]'s Lemma 1 establishes finite-dimensionality from C1-C3; here we additionally require that the discreteness scale appears in the boundary mode count.

(*S4*) *Stinespring/partial-trace structure.* The visible-sector dynamics is generated by partial trace of substratum-level dynamics over the hidden sector, with the trace producing a CPTP channel admitting Stinespring dilation [Main §3.2].

These conditions are weaker than OI's full structural conditions A1–A6: A1 (finiteness) implies S3 with finite ϵ; A6 (background independence) is required for the SM gauge-group derivation but not for the Tier 1 results; A2-A5 (determinism, bounded coupling, center independence, linearity) are required for OI's specific universality-class representative but not for the universality-class-level Tier 1 claims. A substratum satisfying S1-S4 but violating some of A2-A5 (e.g., a substratum with stochastic dynamics, or with nonlinear evolution) would still produce $\hbar = c^3\epsilon^2/(4G)$, the area law with 1/4 coefficient, the CC dissolution, and the Type II RVM functional form, provided the partial-trace machinery and thermal self-consistency hold.

Theorem-level statement. The Tier 1 universality claim can be stated as a theorem at the level of the framework's hierarchical structure (per [Structure §2]):

Theorem (Tier 1 universality). For any substratum $\mathcal{S}$ satisfying S1–S4, the Tier 1 results T1.1–T1.5 follow with the same numerical content as derived in this paper for OI's specific substratum.

The proof is the §3–§7 derivations themselves, with the observation that each step depends only on S1–S4 and not on OI's specific structural conditions A2–A5. A formal proof at the level of explicit theorem-style presentation has not been written out in this paper; the present section establishes the proof's content by identifying the structural conditions required at each step. Writing out the formal theorem explicitly is follow-up work; the universality claim is established at the level of structural-condition identification.

Empirical implication. The Tier 1 universality has substantial empirical content. GW250114 confirms Hawking's area theorem — classical horizon dynamics that any framework retaining general relativity, at any hierarchy level, preserves; it does not measure entropy and therefore does not test the 1/4 coefficient at any level. Conversely, the dark-sector concordance (§7.2), the MOND-like phenomenology (§7.3), and the rotation-curve crossover (§7.3) depend on OI's specific gap-equation solution and are Level D $\times$ Levels G1–G2 predictions; their empirical confirmation would support OI's specific universality-class representative.

The stratification clarifies what would falsify what: a measured violation of the area theorem would challenge the classical horizon dynamics the framework presupposes; a direct empirical test of the 1/4 coefficient, if one becomes available, would test the universality-class claim; a measured deviation from the dark-sector phenomenology would falsify OI's specific representative without necessarily refuting the universality class. The framework's empirical content thus operates at multiple hierarchy levels, with falsifiability stratified accordingly.

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9. Conclusion

The cosmological horizon realizes the embedded-observation framework of [Main], and the consequences for the gravitational sector are sharp. The emergent action scale $\hbar = c^3(2\,l_p)^2/(4G)$ is determined by thermal self-consistency between the classical horizon temperature and the Euclidean periodicity of the emergent QFT, as a gap equation with ϵ the one free geometric input (§4). The Bekenstein-Hawking entropy $S = A/(4l_p^2)$ is derived by direct mode counting, with the 1/4 coefficient fixed as the ratio between the Euclidean KMS period and the Jacobson prefactor — a structural constant, not an empirical fit. The

GW250114 observation of September 2025 confirms Hawking's area theorem at 99.999% confidence — classical horizon dynamics the framework preserves; the derived 1/4 coefficient has no direct empirical test. The cosmological constant problem is dissolved: the 10^{122} ratio between the quantum vacuum energy and the effective Λ is the compression ratio between two levels of description that refer to incommensurable objects, not a failure of cancellation.

The empirical payoff is the set of predictions in §7. The Type II running-vacuum functional form of dark energy is the most distinctive gravitational-sector *structural* prediction: consistent with the Bertini et al. (2025) direct fit to DESI DR2 + DES-Y5 + Planck 2018 [15] (which reports ν consistent with zero at 2σ), and decisively testable by DESI Year 5 and subsequent stage-IV surveys. The coefficient's magnitude at theorem level is the self-consistent emergent-QFT value $|\nu_{\mathrm{QFT}}| \sim 10^{-32}$, observationally indistinguishable from zero. Substratum contributions beyond this are constrained by the boundary mode count of §3.1 and §5 — spatial location alone, with no inherent frequency assignment — to be well below $|\nu_{\mathrm{QFT}}|$ under the natural 2D distribution. The framework's theorem-level prediction is therefore $|\nu| \approx |\nu_{\mathrm{QFT}}|$. The dark-sector corollary explains ~95% of the gravitational budget of the universe without new fundamental particles, and the absence of direct dark-matter detection at the LZ and XENONnT experiments is interpreted within the framework as a structural consequence rather than a null result to be explained away. The MOND phenomenology emerges from entropy displacement at the cosmological boundary with $a_0 = cH/6$, reproducing the baryonic Tully-Fisher relation and predicting Keplerian decline at $r > r_M$ — the latter confirmed by the Jiao et al. (2023) detection of a Keplerian decline in the Milky Way's own rotation curve at ~ 17 kpc. The $H(z)$ scaling of a_0 predicts systematically more baryon-dominated rotation curves at high redshift, consistent with Genzel et al.'s detection of declining outer velocities at $z = 0.9$–2.4 and testable at higher precision by JWST and ELT kinematics.

The foundational framework developed in [Main] and the cubic-lattice realization developed in [SM] are complementary to the present work: [Main] establishes the equivalence between embedded observation and quantum mechanics, the present paper applies it to the cosmological horizon, and the SM paper applies it to the microscopic lattice substratum. The reconstruction theorem and the synthesis claim — that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same deterministic (S, φ) — are developed in [Substratum]. Together, this four-paper synthesis ([Main], [SM], [GR], [Substratum]) establishes that the Standard Model, general relativity, and the arrow of time are three faces of a single finite deterministic construction, with quantum mechanics emerging as the unique self-consistent description for any observer bounded by a causal horizon. The framework's hierarchical structural realism and its relationship to other unification programs are developed in the fifth core paper [Structure].

Appendix A: Nested Partitions

This appendix develops the nested-partition formalism for the framework, establishing the self-consistency of multiple coexisting causal boundaries (typically a black hole horizon inside a cosmological horizon) and deriving the generalized second law and the Page curve as consequences.

A.1 Setup

Let $\mathcal{C}_V$, $\mathcal{C}_B$, $\mathcal{C}_D$ be finite configuration spaces with $|\mathcal{C}_V|, |\mathcal{C}_B| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_B \times \mathcal{C}_D \to \mathcal{C}_V \times \mathcal{C}_B \times \mathcal{C}_D$ be a bijection. The physical picture: V is the region between a black hole horizon and the cosmological horizon, B is the black hole interior, D is the trans-cosmological region. Define $W = V \cup B$ (the cosmological interior).

Two procedures produce a quantum description on V:

Direct: Trace out $B \cup D$ in one step, marginalizing over $\mathcal{C}_B \times \mathcal{C}_D$.

Sequential: (Stage 1) Trace out D, producing emergent QM on W. (Stage 2) Within the emergent QM on W, trace out B, producing a further-reduced description on V.

A.2 Classical Associativity

Theorem A.1. *The direct and sequential procedures produce the same transition probabilities on $\mathcal{C}_V$.*

Proof. The direct transition matrix is:

$$T_{ij}^{\mathrm{dir}}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{b \in \mathcal{C}_B} \sum_{d \in \mathcal{C}_D} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))]$$

The sequential procedure marginalizes over D (Stage 1) then over B (Stage 2). Composing:

$$T_{ij}^{\mathrm{seq}}(t) = \frac{1}{|\mathcal{C}_B|} \sum_{b} \frac{1}{|\mathcal{C}_D|} \sum_{d} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))] = T_{ij}^{\mathrm{dir}}(t)$$

The sums commute (Fubini on a finite product). □

A.3 Quantum-Classical Consistency

Theorem A.2 (Quantum-classical consistency of nested trace-outs). *Let Φ_V^{dir} and Φ_V^{seq} be the CPTP channels on $\mathcal{H}_V$ produced by the direct and sequential procedures respectively. Then $\Phi_V^{dir} = \Phi_V^{seq}$.*

Proof. By the Stinespring dilation theorem [30, 31], φ defines a unitary U_φ on $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_B \otimes \mathcal{H}_D$. The direct channel is $\Phi_V^{\mathrm{dir}}(\rho_V) = \mathrm{Tr}_{BD}[U_\varphi(\rho_V \otimes \rho_B \otimes$

$\rho_D) U_\varphi^\dagger]$. The sequential channel is $\Phi_V^{\rm seq}(\rho_V) = \mathrm{Tr}_B[\mathrm{Tr}_D[U_\varphi\,(\rho_V \otimes \rho_B \otimes \rho_D)\, U_\varphi^\dagger]]$. On finite-dimensional tensor products, $\mathrm{Tr}_B \circ \mathrm{Tr}_D = \mathrm{Tr}_{BD}$, giving $\Phi_V^{\rm seq} = \Phi_V^{\rm dir}$. □

A.4 Hamiltonian and $\hbar$ Consistency

Corollary A.3 (Hamiltonian consistency). *The emergent Hamiltonians $\hat{H}_V^{dir}$ and $\hat{H}_V^{seq}$ satisfy $\hat{H}_V^{dir} = D\,\hat{H}_V^{seq}\,D^\dagger$ for a time-independent diagonal unitary D.*

Proof. Theorem A.2 gives $\Phi_V^{\rm dir} = \Phi_V^{\rm seq}$, hence $T_{ij}^{\rm dir}(t) = T_{ij}^{\rm seq}(t)$ for all t (Born rule applied to diagonal elements). The D-gauge theorem (§3.3) gives the result. □

Corollary A.4 ($\hbar$ universality). *Both procedures yield the same emergent action scale $\hbar = c^3\epsilon^2/(4G)$.*

Proof. $\hbar$ depends only on local boundary quantities c, G, ϵ (§3.2, Steps 2–3), not on which horizon defines the partition or how many trace-outs are performed. □

A.5 Additive Dissipators

Theorem A.5 (Additive dissipators). *Let the classical Hamiltonian be spatially local with coupling chain $V \leftrightarrow B \leftrightarrow D$ (no direct V-D coupling). Let τ_B^D be the D-sector timescale and τ_B^B the B-sector timescale. On observation timescales $t \ll \min(\tau_B^D, \tau_B^B)$, the dissipative correction to unitarity on $\mathcal{H}_V$ decomposes as:*

$$\mathcal{D}_V = \mathcal{D}_V^{(B)} + \mathcal{D}_V^{(D)} + \mathcal{R}$$

where $\|\mathcal{D}_V^{(B)}\| = \mathcal{O}(\tau_S/\tau_B^B)$, $\|\mathcal{D}_V^{(D)}\| = \mathcal{O}(\tau_S/\tau_B^D)$, and the cross term satisfies $\|\mathcal{R}\| = \mathcal{O}(\tau_S^2/(\tau_B^B \cdot \tau_B^D))$.

Proof. The proof applies the boundary-only dependence lemma (§3.2) twice, tracking the cross term at each stage.

(i) *First application: freeze D.* Apply the boundary-only dependence lemma to the partition (W, D) where $W = V \cup B$. By spatial locality, $H_{\rm tot} = H_V + H_B + H_D + H_{VB} + H_{BD}$ with no direct V-D coupling. The D-sector is frozen on timescales $t \ll \tau_B^D$, giving:

$$T_{ij}^V(t) = \frac{1}{|\mathcal{C}_B|} \sum_{b \in \mathcal{C}_B} \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i, b))] + \mathcal{O}(t/\tau_B^D)$$

where φ_t^{VB} is the flow restricted to $V \times B$ with D frozen. The $\mathcal{O}(t/\tau_B^D)$ correction arises from D's back-reaction on B through H_{BD}.

(ii) *Dissipator from the frozen-D dynamics.* The frozen-D flow φ_t^{VB} is a bijection on $\mathcal{C}_V \times \mathcal{C}_B$ — the standard Stinespring setup with B as the hidden sector. The emergent dynamics on V is approximately unitary with dissipator $\mathcal{D}_V^{(B)} \sim \mathcal{O}(\tau_S/\tau_B^B)$.

(iii) *Decomposition of the D-correction.* The $\mathcal{O}(t/\tau_B^D)$ correction from step (i) has two components:

Direct: D's back-reaction shifts the boundary modes of B by $\delta b \sim \mathcal{O}(t/\tau_B^D)$. This propagates through H_{VB} to the visible sector, contributing $\mathcal{D}_V^{(D)} \sim \mathcal{O}(\tau_S/\tau_B^D)$.

Cross: The shift δb also modifies the effective B-state against which $\mathcal{D}_V^{(B)}$ is computed. The dissipator $\mathcal{D}_V^{(B)}$ depends on the B-state through the Liouville average over b. The D-induced shift changes this average by $\mathcal{O}(t/\tau_B^D)$, so:

$$\|\mathcal{R}\| = \left\|\frac{\partial \mathcal{D}_V^{(B)}}{\partial b}\right\| \cdot \|\delta b\| \leq \|\mathcal{D}_V^{(B)}\| \cdot \mathcal{O}(t/\tau_B^D) = \mathcal{O}\left(\frac{\tau_S}{\tau_B^B} \cdot \frac{\tau_S}{\tau_B^D}\right) = \mathcal{O}\left(\frac{\tau_S^2}{\tau_B^B \cdot \tau_B^D}\right)$$

The inequality uses the fact that $\mathcal{D}_V^{(B)}$ is a continuous function of the B-state distribution and the shift is first-order in t/τ_B^D. $\square$

Bound for the cosmological + stellar black hole case. For a $30M_\odot$ black hole inside the cosmological horizon: $\tau_S \sim 10^{-15}$ s, $\tau_B^B \sim \tau_{\text{scr}} \sim \beta \log S_{\text{BH}} \sim 10^{-3}$ s, $\tau_B^D \sim 1/H \sim 10^{17}$ s. The three terms:

$$\|\mathcal{D}_V^{(B)}\| \sim 10^{-12}, \qquad \|\mathcal{D}_V^{(D)}\| \sim 10^{-32}, \qquad \|\mathcal{R}\| \sim 10^{-44}$$

The cross term is negligible by 12 orders of magnitude relative to the smaller of the two primary dissipators.

A.6 The Generalized Second Law

The Bekenstein-Hawking entropy of a horizon is the information content of the corresponding partition boundary (§5): $S_{\text{BH}} = A_{\text{BH}}/(4\, l_p^2)$ modes at the black hole horizon, $S_{\text{dS}} = A_{\text{dS}}/(4\, l_p^2)$ modes at the cosmological horizon. The matter entropy is the von Neumann entropy of the visible-sector state: $S_{\text{matter}} = -\text{Tr}[\rho_V \log \rho_V]$.

Theorem A.6 (Generalized second law). *For any process described by the CPTP channel Φ_V on $\mathcal{H}_V$:*

$$S_{\text{matter}}(t) + S_{\text{BH}}(t) + S_{\text{dS}}(t) \geq S_{\text{matter}}(0) + S_{\text{BH}}(0) + S_{\text{dS}}(0)$$

Proof. The total system (S, φ) evolves unitarily at the fundamental level, so the total fine-grained entropy $S_{\text{total}} = \log |S|$ is constant. The three terms S_{matter}, S_{BH}, S_{dS} are the observer's ignorance about three sectors: V, B, D respectively.

(i) *Subadditivity.* For the tripartite system $\mathcal{H}_V \otimes \mathcal{H}_B \otimes \mathcal{H}_D$ in a pure state $|\Psi\rangle$ (the total state is deterministic, hence pure conditioned on the initial state), the strong subadditivity of von Neumann entropy [31] gives:

$$S(\rho_{VB}) + S(\rho_{BD}) \geq S(\rho_B) + S(\rho_{VBD})$$

Since $\rho_{VBD} = |\Psi\rangle\langle\Psi|$ is pure, $S(\rho_{VBD}) = 0$. Thus $S(\rho_{VB}) + S(\rho_{BD}) \geq S(\rho_B)$.

(ii) *Identification.* The observer's matter entropy is $S_{\text{matter}} = S(\rho_V)$. By purification on the tripartite system: $S(\rho_V) = S(\rho_{BD})$. The black hole entropy is $S_{\text{BH}} = S(\rho_B)$ restricted to the boundary modes (the number of B-modes coupled across the black hole horizon). The cosmological entropy is $S_{\text{dS}} = S(\rho_D)$ restricted to the boundary modes.

(iii) *Monotonicity and compensation.* The hidden-sector prior is uniform (Liouville marginalization), so the marginal channel is unital: $\Phi_V(I_V) = \text{Tr}_{BD}[U_\varphi\,(I_V \otimes \rho_B \otimes \rho_D)\,U_\varphi^\dagger] = I_V$ for maximally mixed ρ_B, ρ_D. Unital channels are entropy-non-decreasing (their outputs are majorized by their inputs [31]), so $S(\rho_V)$ is non-decreasing under the emergent dynamics. (Relative-entropy contraction alone would not suffice — general CPTP maps can decrease entropy; the uniform prior is load-bearing here.) Any process that decreases S_{BH} (shrinking the black hole) transfers information from B to $V \cup D$. By the unitarity of φ, the total mutual information $I(V : B : D)$ is conserved, so the decrease in S_{BH} is compensated by an increase in $S_{\text{matter}} + S_{\text{dS}}$. $\square$

A.7 The Page Curve

Setup. A black hole with initial horizon area A_0 evaporates. The partition boundary at the black hole horizon moves: modes transition from B (interior) to R (radiation, a subsystem of V) as the horizon shrinks. Define $n_B(t) = A_{\text{BH}}(t)/(4\,l_p^2)$ (the number of boundary modes at the BH horizon) and $n_R(t) = n_B(0) - n_B(t)$ (the number of emitted radiation modes). Conservation: $n_B(t) + n_R(t) = n_B(0) = S_{\text{BH}}(0)$.

The trans-cosmological sector D is frozen on evaporation timescales ($\tau_{\text{evap}} \ll \tau_B^D$), so by Theorem A.5 the relevant dynamics is the bipartite system $B \cup R$ with D decoupled at leading order.

Theorem A.7 (Effective bipartite pure state). *On timescales $t \ll \tau_B^D$, the joint state of B and R is effectively pure: $\rho_{BR} = |\psi(t)\rangle\langle\psi(t)| + \mathcal{O}(\tau_S/\tau_B^D)$, where $|\psi(t)\rangle \in \mathcal{H}_B(t) \otimes \mathcal{H}_R(t)$.*

Proof. The total system evolves unitarily under U_φ (§A.3). The total state $|\Psi\rangle \in \mathcal{H}_V \otimes \mathcal{H}_B \otimes \mathcal{H}_D$ is pure (deterministic evolution from a definite initial state). By the boundary-only dependence lemma, the D-sector is frozen: ρ_D is approximately constant, and $|\Psi\rangle \approx |\psi_{VB}\rangle \otimes |\chi_D\rangle + \mathcal{O}(\tau_S/\tau_B^D)$. Restricting to the $B \cup R$ subsystem (where $R \subset V$ is the radiation): $\rho_{BR} \approx |\psi\rangle\langle\psi|$ is pure at leading order. The correction is $\mathcal{O}(\tau_S/\tau_B^D) \sim 10^{-32}$. $\square$

Theorem A.8 (Cycle typicality). *For the dynamics φ on the energy shell $\Sigma_E \subset \mathcal{H}_B \otimes \mathcal{H}_R$, the time-averaged entanglement entropy of the radiation equals the Page value [32]:*

$$\overline{S(\rho_R)} = S_{\text{Page}}(d_B, d_R) + \mathcal{O}(\epsilon)$$

for all initial conditions except a fraction $\leq (\ln d_{\min}/\epsilon)\, e^{-c\, d_{\min}\, \epsilon^2}$, where $d_B = |\mathcal{C}_B(t)|$, $d_R = |\mathcal{C}_R(t)|$, $d_{\min} = \min(d_B, d_R)$, and:

$$S_{\text{Page}}(d_B, d_R) = \ln(\min(d_B, d_R)) - \frac{\min(d_B, d_R)}{2\max(d_B, d_R)}$$

Proof. The bijection φ conserves energy (the wave equation is Hamiltonian), so φ restricts to a bijection on each energy shell Σ_E. Since Σ_E is finite, this restricted bijection decomposes into disjoint cycles $C_1, \dots, C_k$ with lengths $L_1, \dots, L_k$ summing to $|\Sigma_E|$. On each cycle C_j, the time average of any observable f is exact:

$$\overline{f}_{C_j} = \frac{1}{L_j} \sum_{s \in C_j} f(s)$$

By Popescu-Short-Winter [29], the set of atypical states — those with $|S(\mathrm{Tr}_B|\psi\rangle\langle\psi|) - S_{\text{Page}}| > \epsilon$ — has measure at most $\eta = e^{-c\, d_{\min}\, \epsilon^2}$ on Σ_E, where $d_{\min} = \min(d_B, d_R)$ and $c > 0$ is a universal constant. The total number of atypical states is at most $\eta|\Sigma_E|$.

Call a cycle *bad* if its time-averaged entropy deviates from S_{Page} by more than 2ϵ. Every state's deviation is bounded by $\ln d_{\min}$, so a cycle with atypical fraction f_j satisfies $|\overline{S}_{C_j} - S_{\text{Page}}| \leq \epsilon + f_j \ln d_{\min}$; a bad cycle therefore requires $f_j \geq \epsilon / \ln d_{\min}$, i.e. $L_j \leq (\ln d_{\min}/\epsilon)\, |\{s \in C_j : s \text{ atypical}\}|$. Summing over all bad cycles:

$$\sum_{j\,\text{bad}} L_j \leq \frac{\ln d_{\min}}{\epsilon}\, \eta\, |\Sigma_E|$$

The fraction of initial conditions on Σ_E that land on a bad cycle is therefore at most $(\ln d_{\min}/\epsilon)\, \eta = (\ln d_{\min}/\epsilon)\, e^{-c\, d_{\min}\, \epsilon^2}$ — the polynomial prefactor is negligible against the exponential concentration. For a $30 M_\odot$ black hole, $d_{\min} \sim e^{S_{\text{BH}}} \sim e^{10^{77}}$, so $\eta \sim e^{-10^{77}}$. For all but an exponentially negligible fraction of initial conditions, the time-averaged entanglement entropy equals the Page value. $\square$

Theorem A.9 (The Page curve). *The entanglement entropy of the radiation follows:*

$$S_{\text{rad}}(t) = \begin{cases} n_R(t) \cdot \ln q - \frac{q^{n_R(t)}}{2\,q^{n_B(t)}} & \text{for } n_R(t) < n_B(t) \quad \text{(early: before Page time)} \\ n_B(t) \cdot \ln q - \frac{q^{n_B(t)}}{2\,q^{n_R(t)}} & \text{for } n_R(t) > n_B(t) \quad \text{(late: after Page time)} \end{cases}$$

where q is the alphabet size (gauge — it cancels in all ratios), $n_B(t) + n_R(t) = n_B(0)$, and the Page time t_P is defined by $n_R(t_P) = n_B(t_P) = n_B(0)/2$.

Proof. Direct substitution of $d_B = q^{n_B(t)}$ and $d_R = q^{n_R(t)}$ into the Page entropy formula (Theorem A.8), using $\ln(q^n) = n \ln q$ and the conservation law $n_B + n_R = n_B(0)$. The turnover at $n_R = n_B$ follows from $\min(d_B, d_R)$ switching from d_R to d_B. □

The evaporation trajectory. The Hawking temperature (§3.2, Step 4) gives luminosity $L \sim 1/M^2$, hence $dM/dt = -\alpha/M^2$ where $\alpha = \hbar c^4 \Gamma/(15360\pi G^2)$ and Γ is the species-dependent greybody factor (solution-specific). Integrating:

$$M(t) = M_0\big(1 - t/t_{\text{evap}}\big)^{1/3}, \qquad t_{\text{evap}} = \frac{M_0^3}{3\alpha}$$

The boundary mode count $n_B(t) = 4\pi G M(t)^2/(\hbar c) = n_B(0) \cdot (1 - t/t_{\text{evap}})^{2/3}$ and $n_R(t) = n_B(0) - n_B(t)$. The Page time t_P satisfies $n_B(t_P) = n_B(0)/2$:

$$t_P = t_{\text{evap}}\big(1 - 2^{-3/2}\big) \approx 0.646\, t_{\text{evap}}$$

The entropy of the radiation at the Page time is $S_{\text{rad}}(t_P) = n_B(0) \ln q/2 = S_{\text{BH}}(0) \ln q/2$ — half the initial black hole entropy, as expected.

Remark (Comparison with QES + islands). The Page-curve derivation above quantitatively matches the prediction obtained via the quantum extremal surface formula and the entanglement-island construction [41, 42, 43, 44]. In the QES approach, the Page curve emerges through an extremization principle that selects the dominant saddle in a replica-trick computation, with the post-Page-time saddle including an "island" — a region of bulk spacetime that contributes to the radiation's entanglement entropy despite being topologically disconnected. The replica wormhole geometries that connect the replicas in the gravitational path integral are essential to obtaining the saturating Page curve at late times.

The framework here arrives at the same Page time $t_P = (1 - 2^{-3/2}) t_{\text{evap}}$ and the same Page entropy $S_{\text{rad}}(t_P) = S_{\text{BH}}(0) \ln q/2$ through a structurally cleaner mechanism: the cycle-typicality argument of Theorem A.8 invokes only Popescu-Short-Winter [29] applied to the deterministic substratum dynamics on the energy shell, with no replica trick, no ergodicity assumption, and no gravitational path integral. The "island" in the QES formulation corresponds, in the present construction, to the post-Page-time regime in which the BH boundary modes form the smaller subsystem; the "replica wormhole" geometries appear to correspond to the substratum's cycle structure on the energy shell. This suggested correspondence — replica wormholes substratum cycles — is conjectural at this stage; making it precise would require constructing the gravitational path integral as an effective description of the substratum dynamics in the appropriate limit. What the present derivation establishes unconditionally is that the Page curve's quantitative content is reproducible from a finite deterministic substratum without invoking the replica machinery.

The convergence is structural: both approaches require *something* to play the role of typicality (PSW measure-concentration here, replica saddles in QES + islands), and both extract the same Page time and Page entropy from that ingredient. The framework here establishes that the typicality-providing ingredient need not be gravitational at all — it can be combinatorial, on the substratum side of the trace-out — which constitutes a microscopic realization of the entanglement-island construction.

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